Human Factors

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Overview

1. Perception
   - Biases
   - Structure
   - Limitations
   - HCI and vision
   - Design implications
2. Memory
   - Design implications
3. Personality
4. Motivation
5. Cognition
   - Attention
   - Learning
   - Reasoning
   - Design implications

Sources

- Jeff, **Designing with the Mind in Mind: Simple Guide to Understanding User Interface Design Guidelines**, 2nd ed. (2014).
- Mackenzie, Chapter 2, **Human Factors**, Human Computer Interaction: An Empirical Research Perspective, 1st ed. (2013)

HCI challenge

- **Human variability** is the biggest challenge of HCI
- Obviously, understanding humans increase our success
- Here, we discuss human **perception**, **memory** and **cognition**

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Perception

- **First stage of processing** in the brain, occurs when sensory signals are received as input. It is at this stage human makes meanings
- Perception has been studied in a area of experimental psychology known as **psychophysics** - examines the relationship between perception and physical phenomena

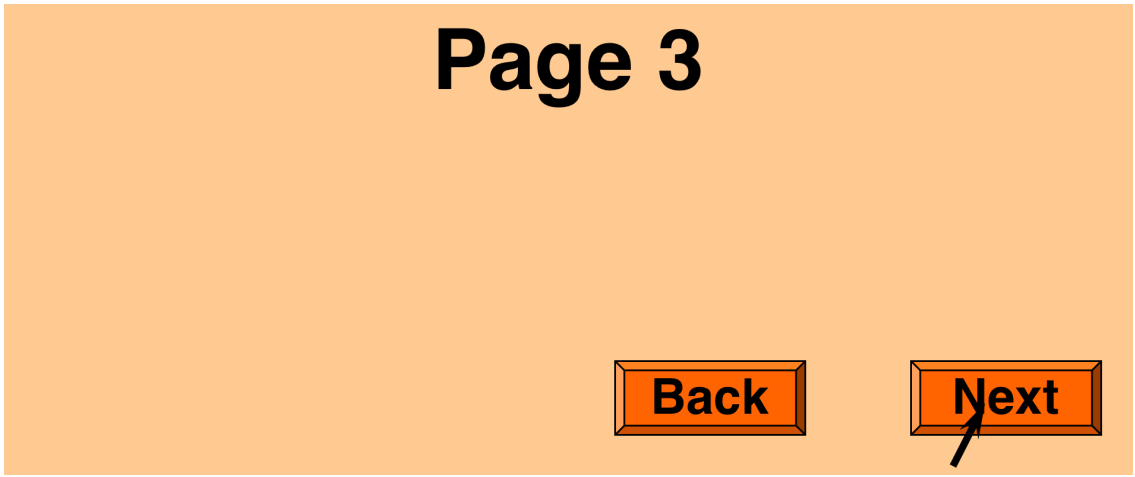
Facts about Perception

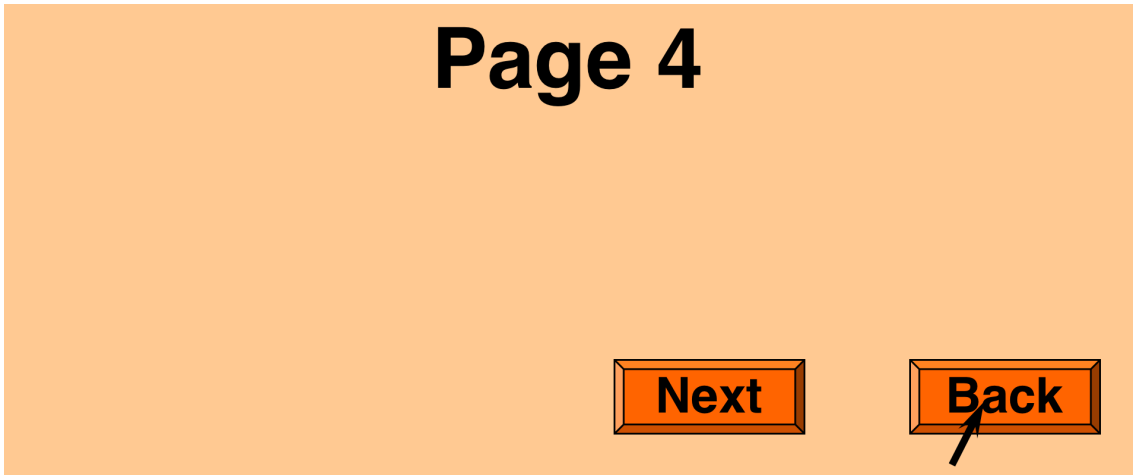
1. Our perception is biased by
   - our goals
   - our belief
   - our experience
   - the context
2. Our vision is optimized to see structure
3. Our color vision is limited
4. Our peripheral vision is poor
5. Visual search is linear unless target "pops"
6. Reading is unnatural

Our perception is biased...



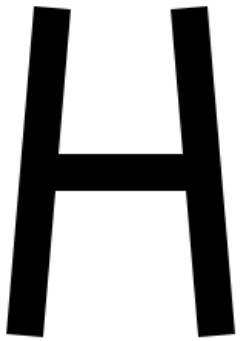
Our perception is biased...

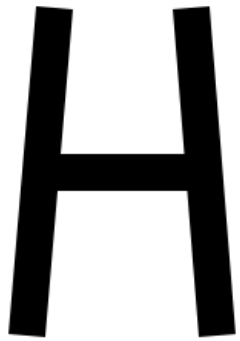




Our perception is biased...

Exactly same character





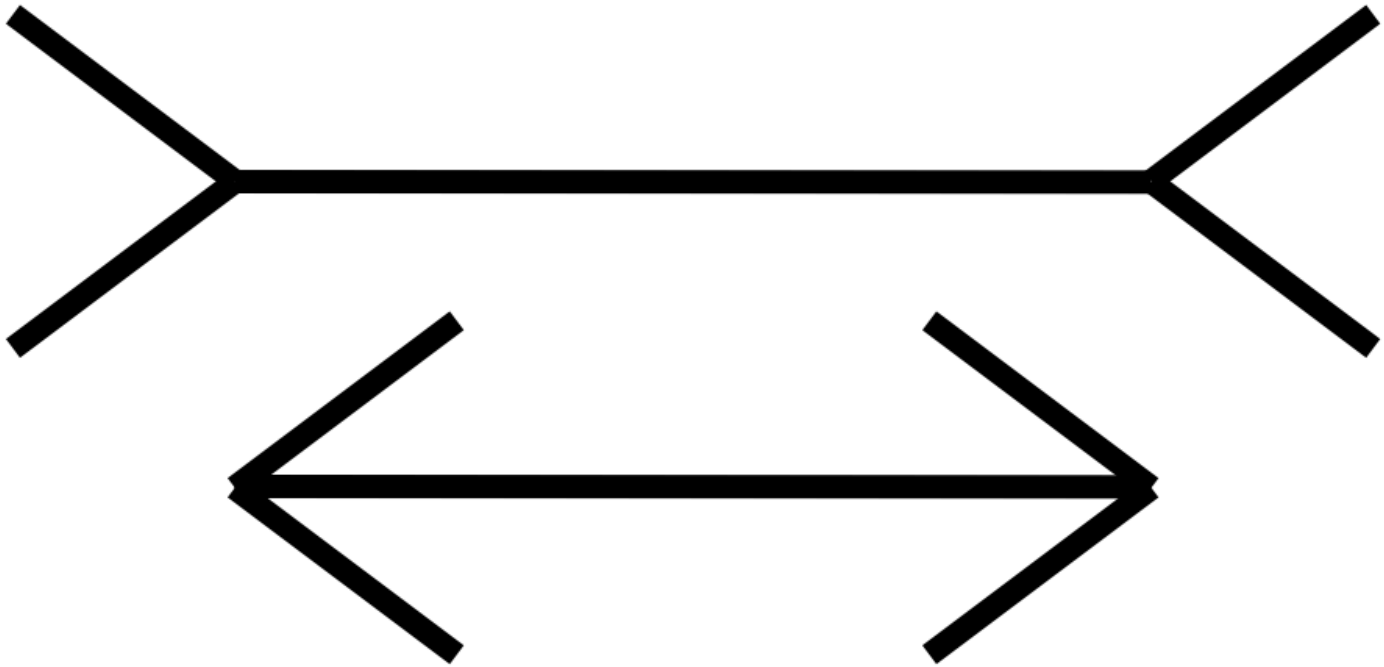
Our perception is biased...

But our perception can be changed based on the **context**



Our perception is biased...

Muller-Lyer illusion



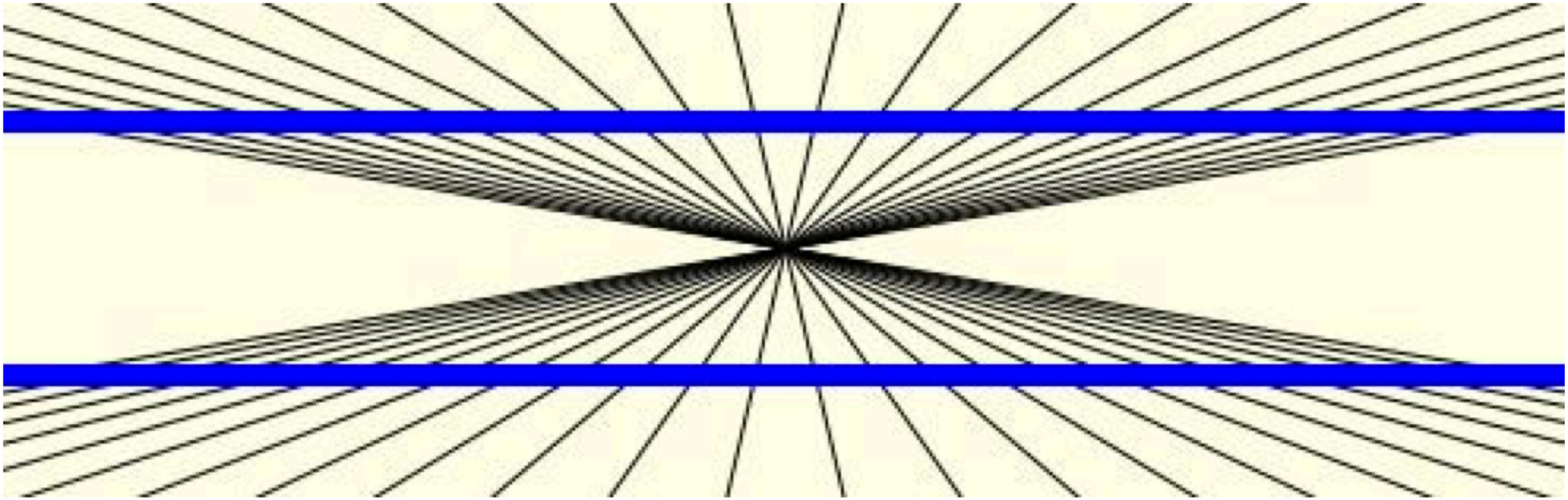
Our perception is biased...

Muller-Lyer illusion



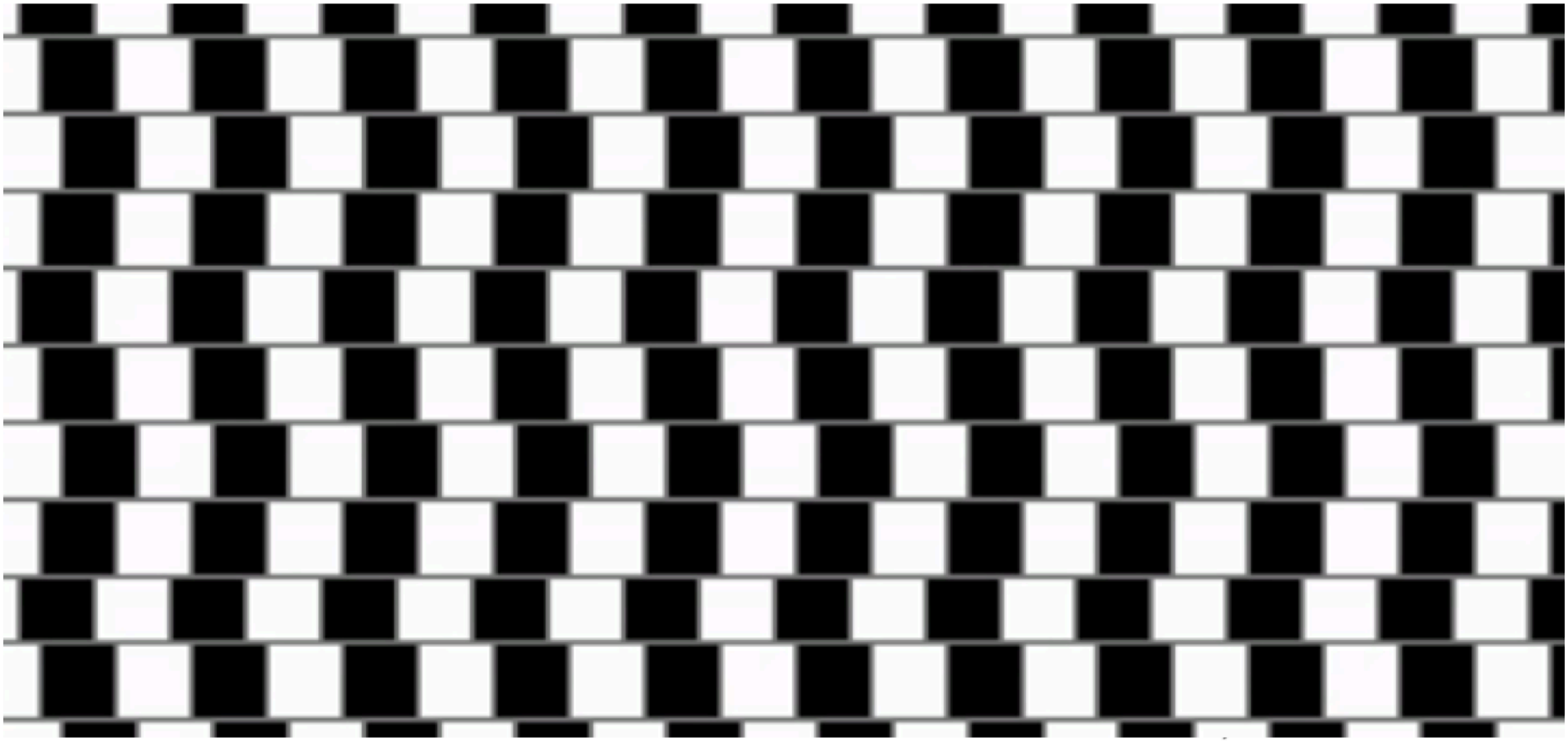
Our perception is biased...

All lines in this image are straight!



Our perception is biased...

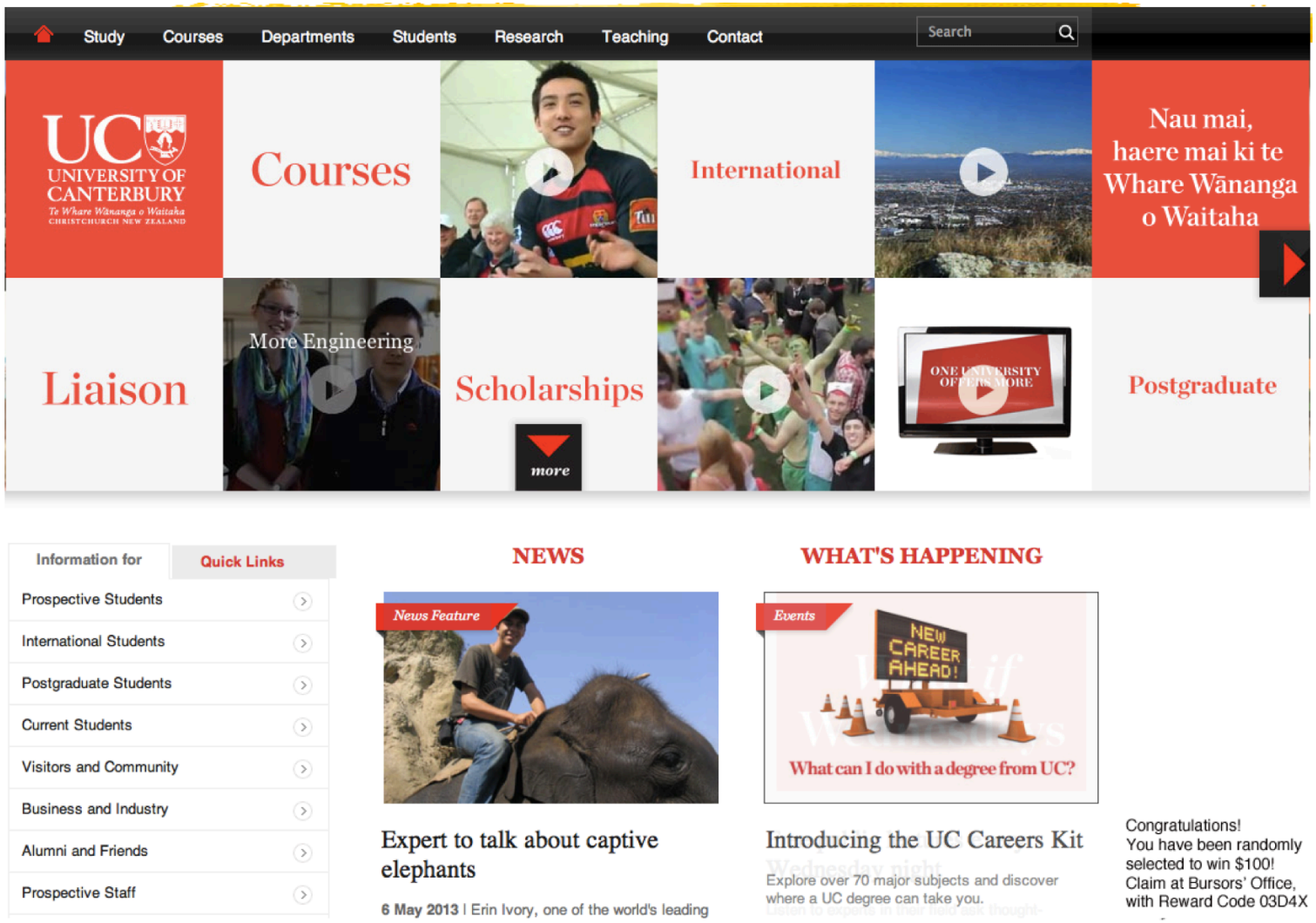
Gray lines are straight, horizontal, and parallel!



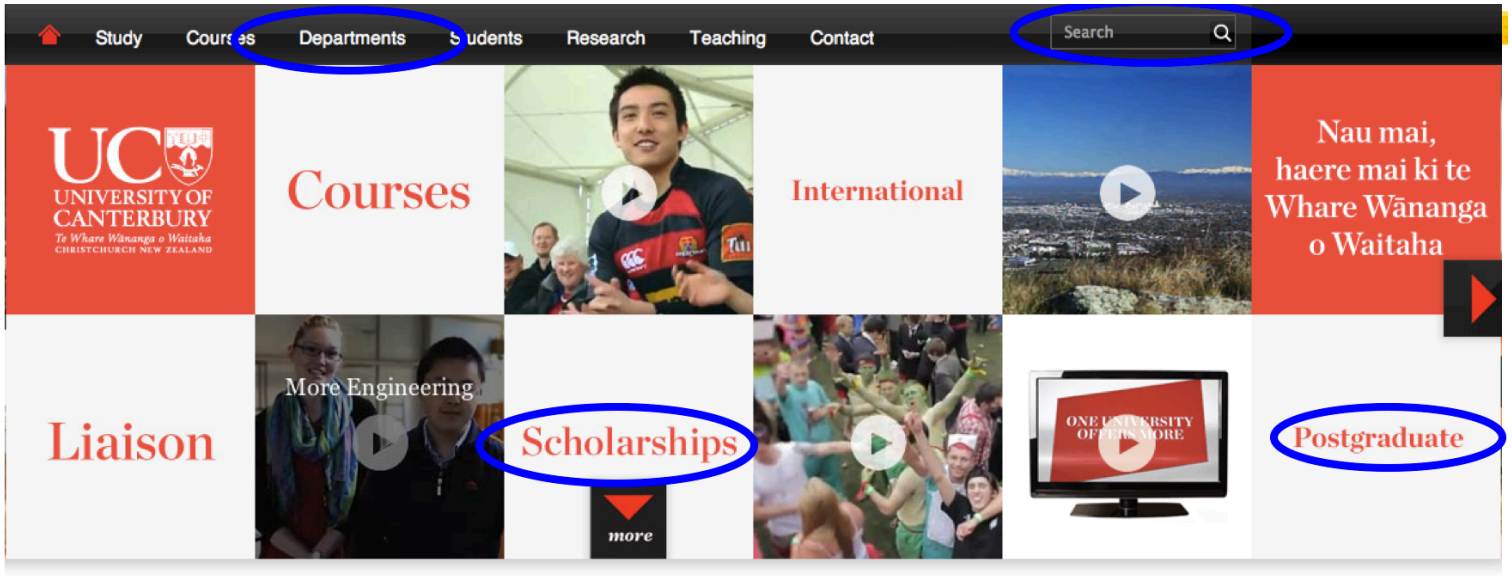
Our perception is biased by goals

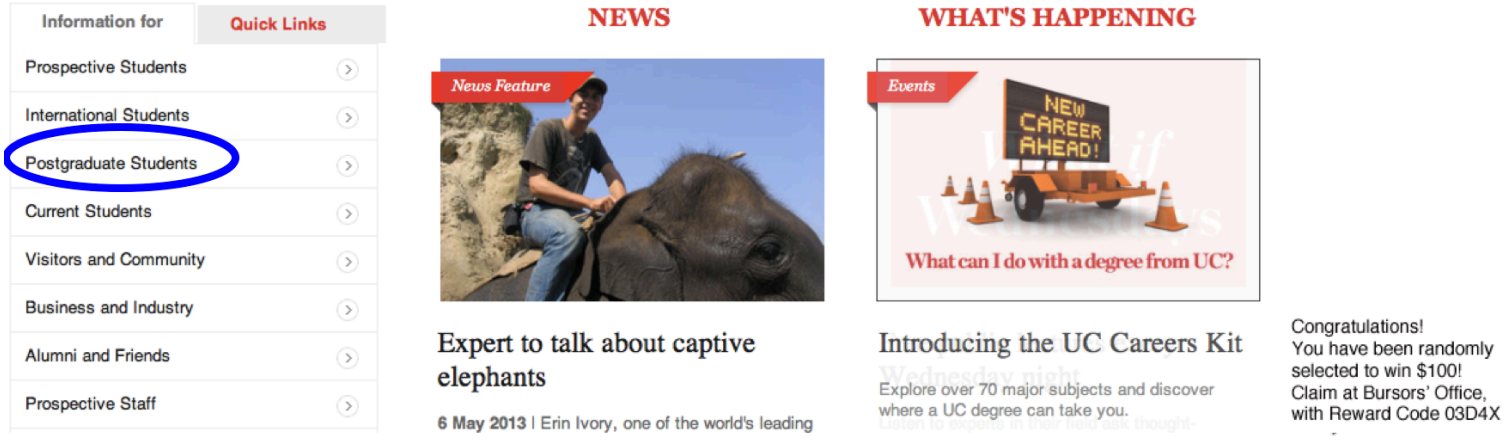
- Our perception is biased toward our **goals**
- Tend not to notice things unrelated to goals

Our perception is biased by goals



Our perception is biased by goals





Our vision is optimized to see structures

Gestalt Principles of Visual Perception

- Proximity
- Similarity
- Continuity
- Closure
- Symmetry
- Figure/ground
- Common fate

Proximity

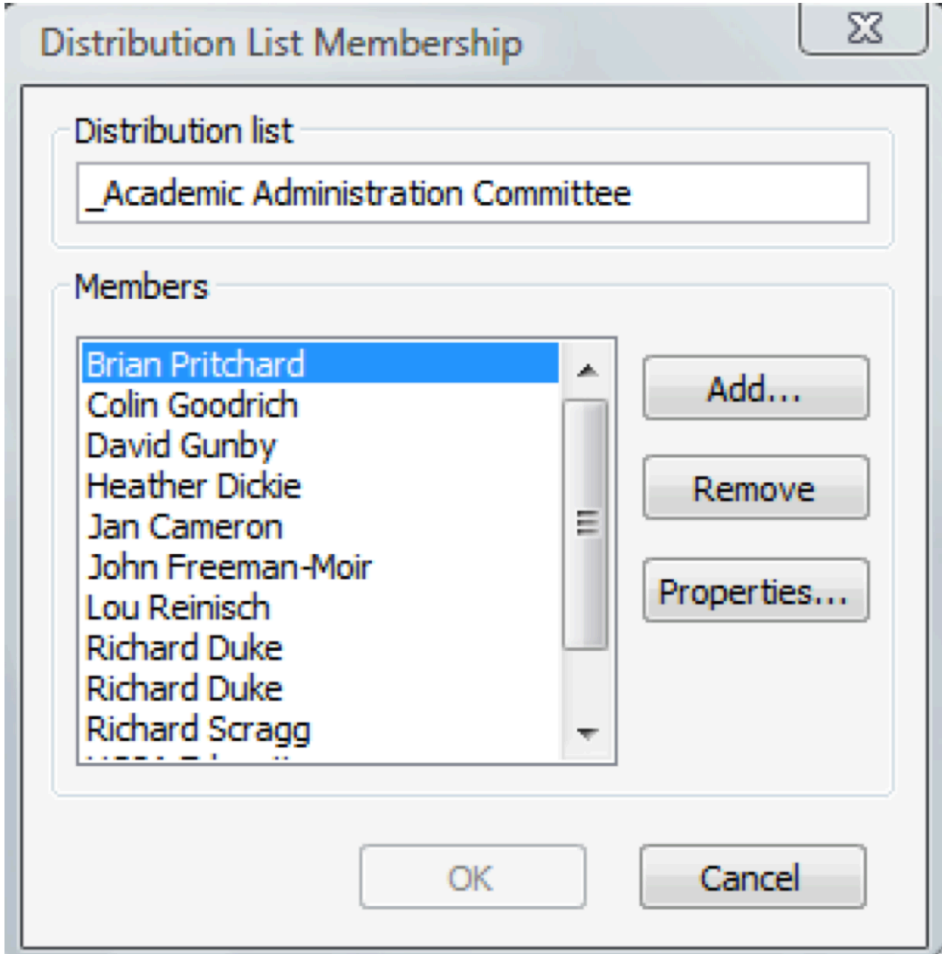
Items that are closer appear grouped



Left: rows

Right: columns

Proximity


Distribution List Membership
Distribution list
_Academic Administration Committee
Members
Brian Pritchard
Colin Goodrich
David Gunby
Heather Dickie
Jan Cameron
John Freeman-Moir
Lou Reinisch
Richard Duke
Richard Duke
Richard Scragg
Add...
Remove
Properties...
OK
Cancel

Proximity

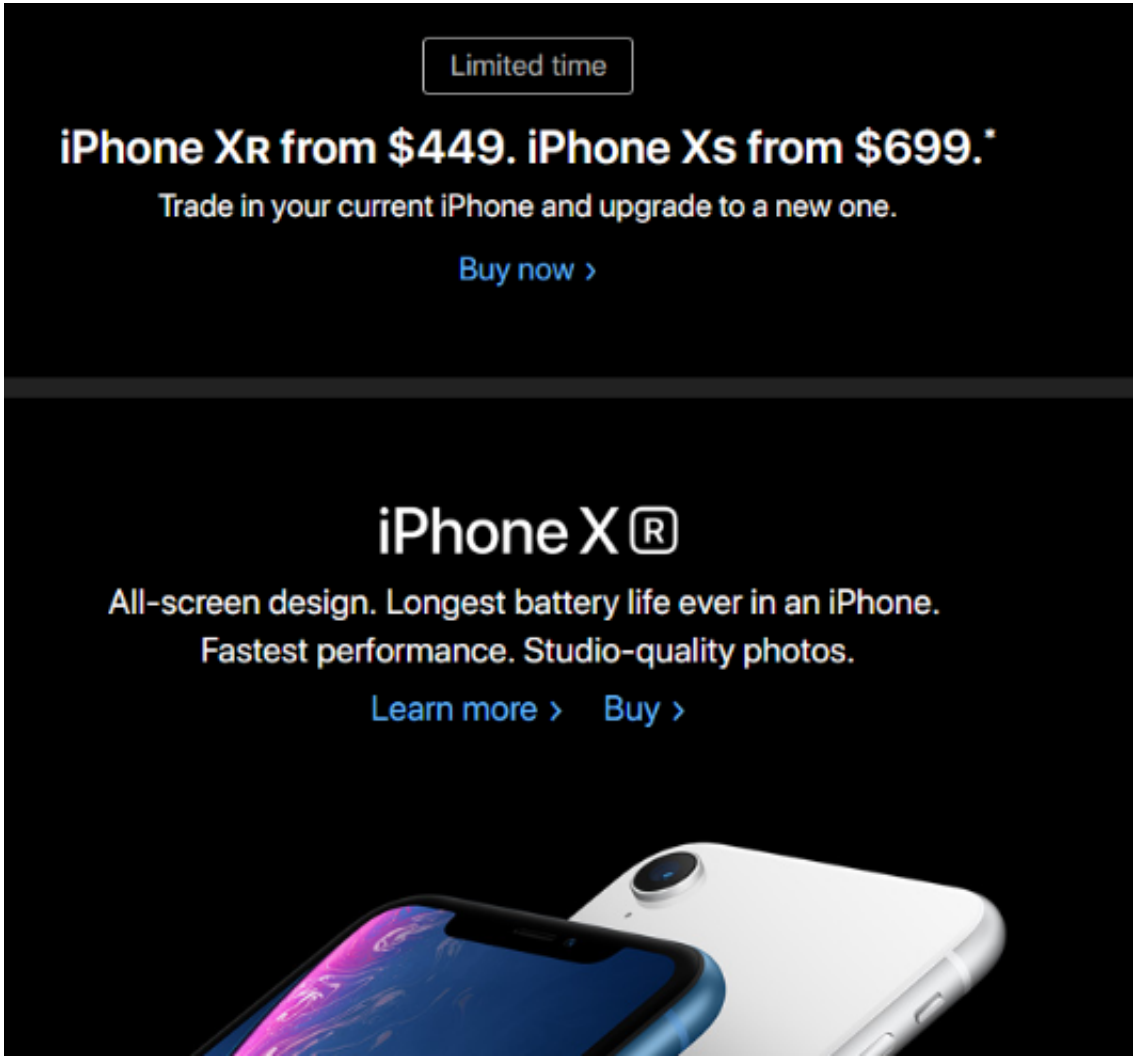
Orchidaceae - Wikipedia, the free encyclopedia
en.wikipedia.org/wiki/**Orchid**aceae
Orchids are easily distinguished from other plants, as they share some very evident apomorphies. Among these are: bilateral symmetry (zygomorphism), many ...
Etymology - Distribution - Taxonomy - Characteristics

orchids, gift **orchids**, hobby growing
www.**orchids**.com/
Orchids.com Logo, Login | Home | My Account | Customer Service · Sign In or Register | Shopping Cart. 1-888-4ORCHID (1-888-467-2443) - 9 am-5 pm (Pacific) ...

Pacific **Orchid** Exposition - San Francisco **Orchid** Society
www.**orchids**anfrancisco.org/poe.html
The San Francisco **Orchid** Society would like to express its thanks to the Taiwan Tourist Bureau CBS and KCBS for their continued support in production and ...

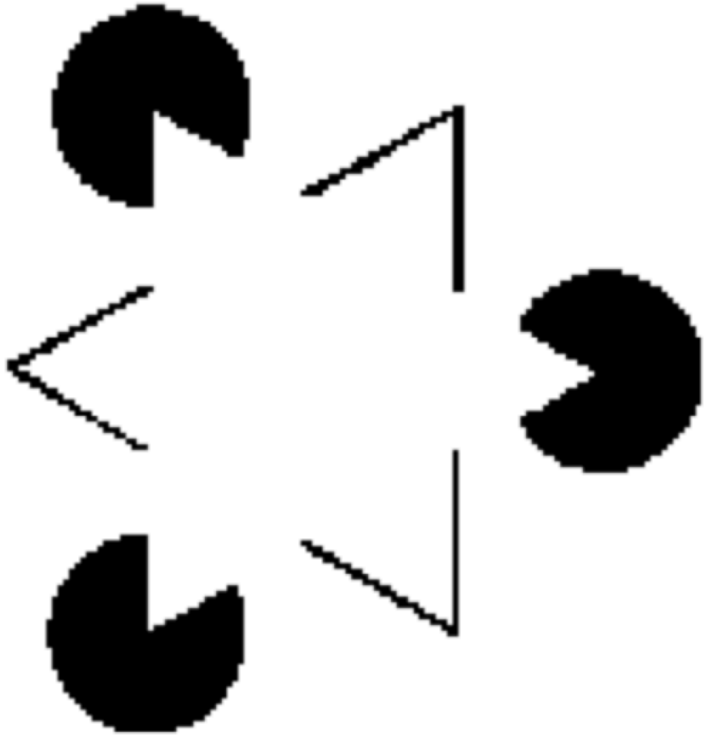
Orchids in the Park - San Francisco **Orchid** Society
www.**orchids**anfrancisco.org/**orchids**inthepark.html
Thanks to all volunteers and attendees for a successful "**Orchids** In The Park" event. -- Images by Jeff Harris. --Images by Jeff Harris. <back to top> ...

Proximity


Limited time
iPhone XR from $449. iPhone Xs from $699.*
Trade in your current iPhone and upgrade to a new one.
Buy now ›
iPhone XR
All-screen design. Longest battery life ever in an iPhone.
Fastest performance. Studio-quality photos.
Learn more ›
Buy ›

Closure

We tend to see whole, closed objects, not collections of fragments



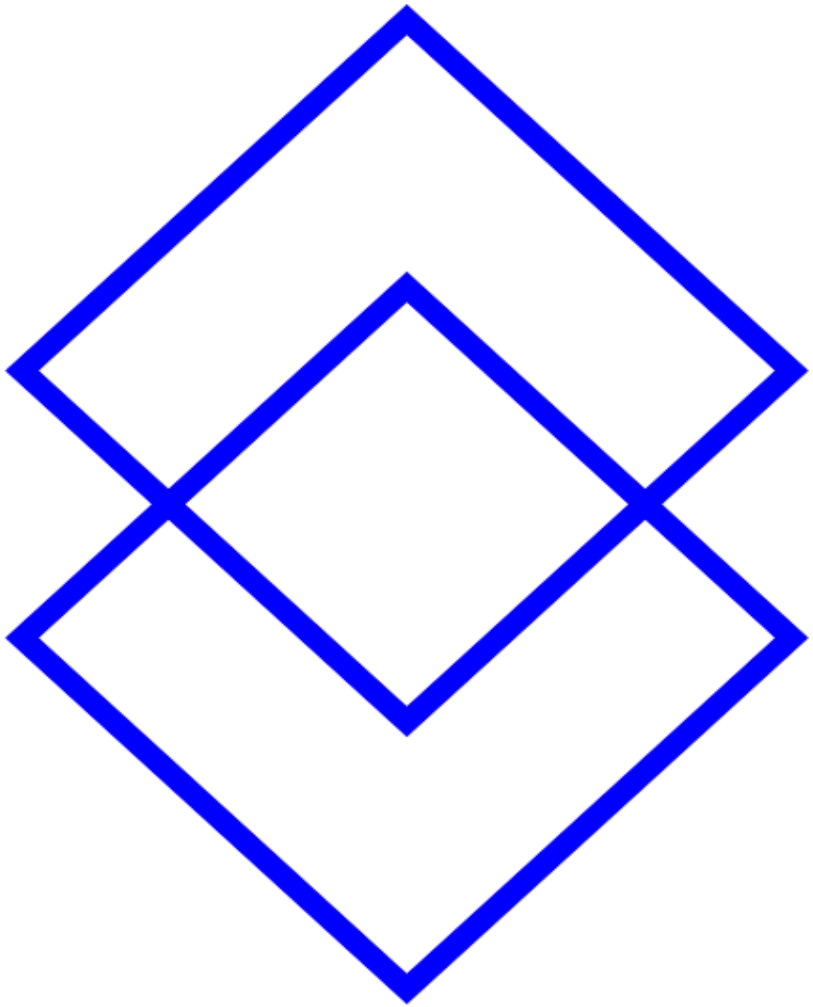
Closure

We tend to see whole, closed objects, not collections of fragments



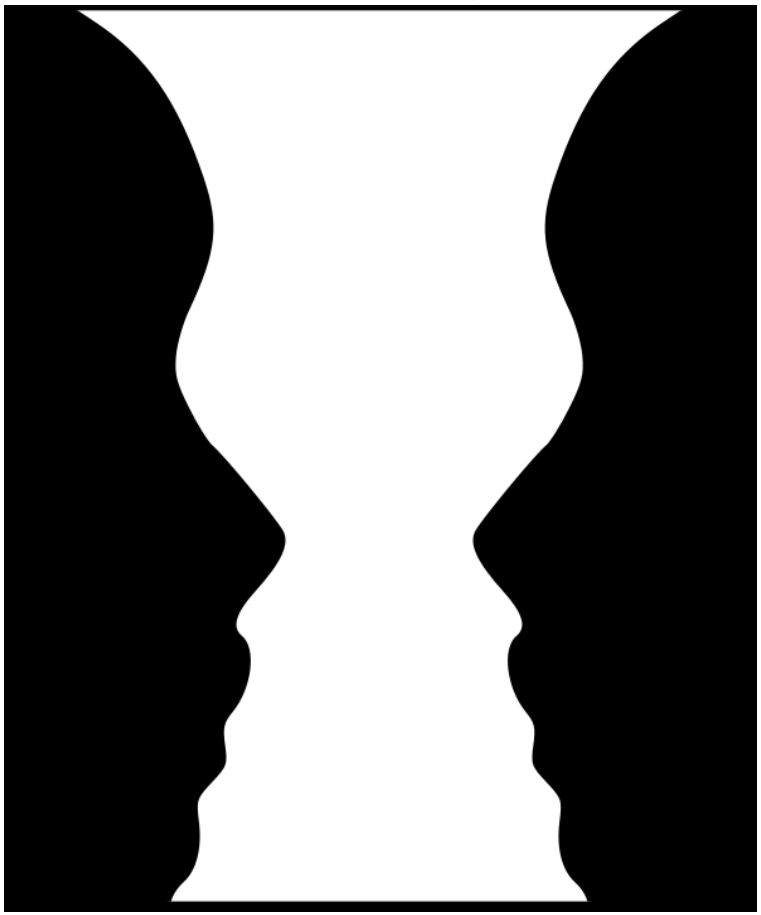
Simplicity

We tend to see simple figures rather than complex ones



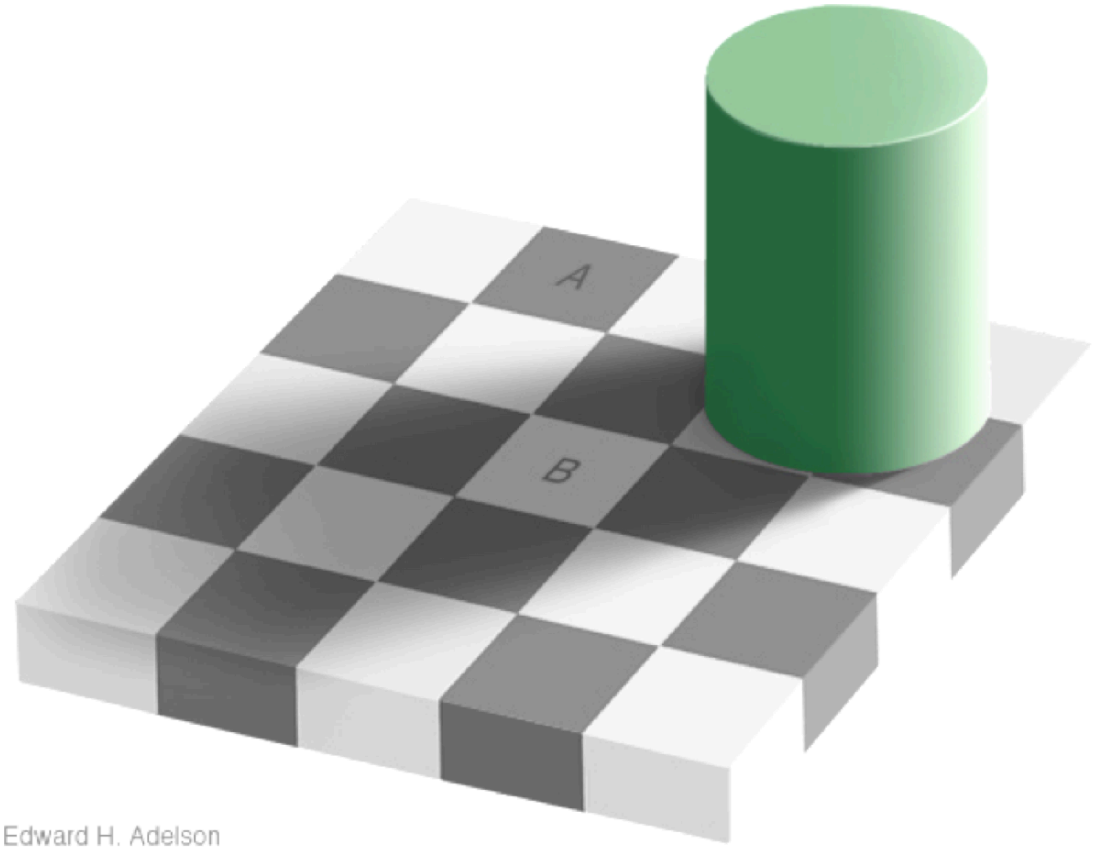
Figure/Ground

The tendency to simplify based on the figures and the grounds



Our color vision is limited...

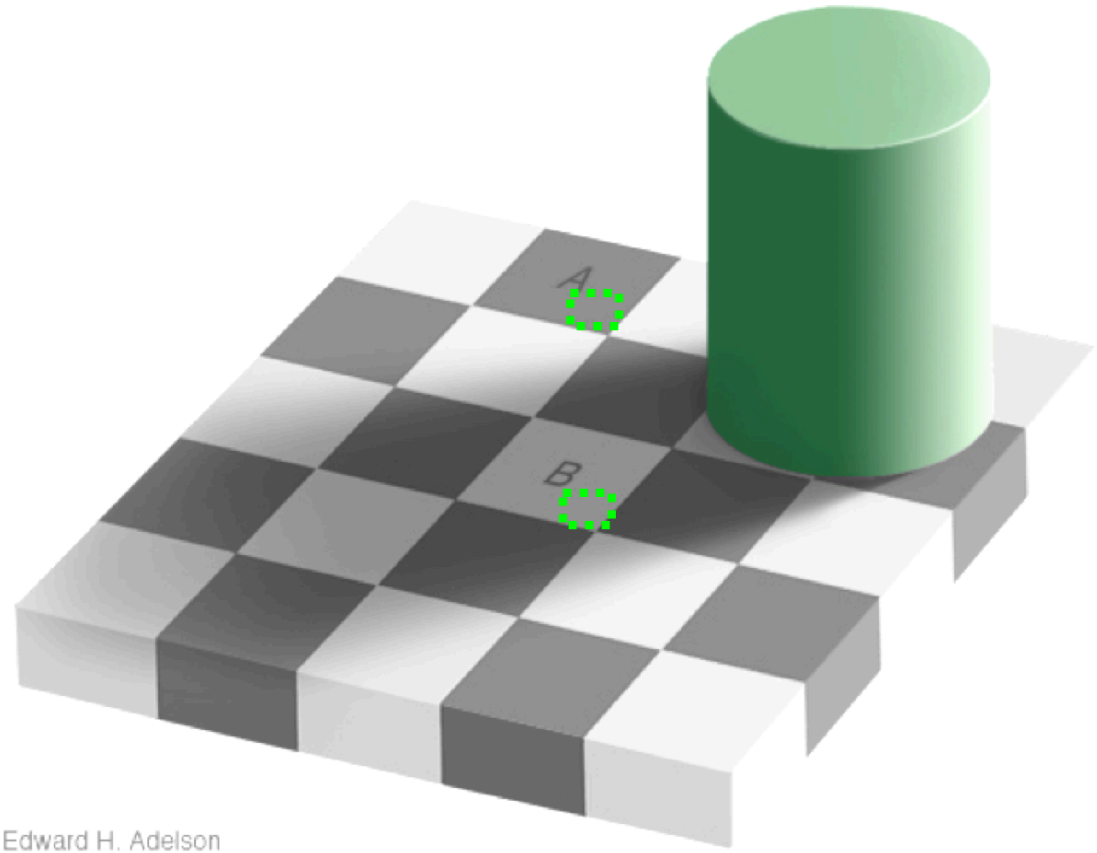
Optimized to see contrasts, edges, and changes, not absolute levels



Edward H. Adelson

Our color vision is limited...

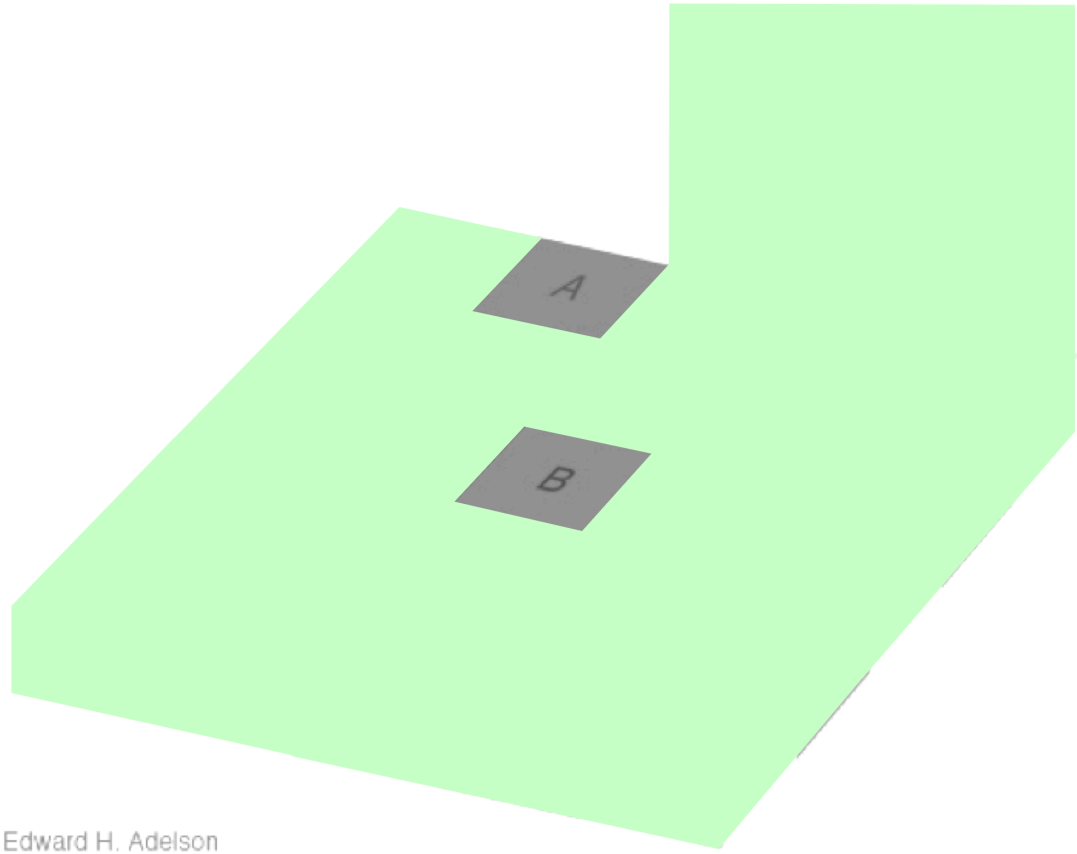
Optimized to see contrasts, edges, and changes, not absolute levels



Edward H. Adelson

Our color vision is limited...

Optimized to see contrasts, edges, and changes, not absolute levels



Edward H. Adelson

Our color vision is limited...

We have trouble discriminating - pale colors, small color patches, separated patches...



Our color vision is limited...

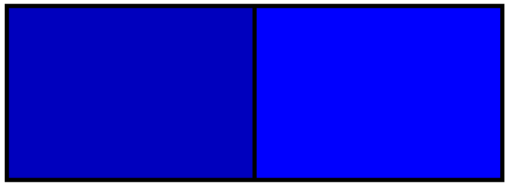
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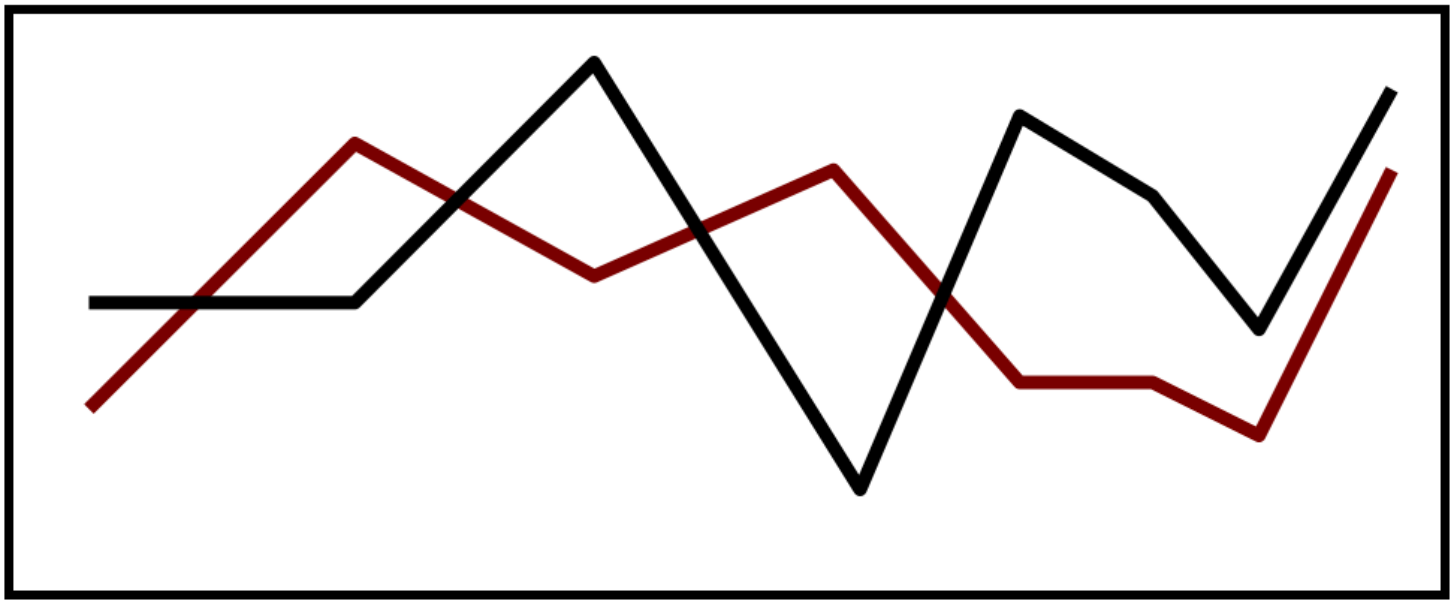


Our color vision is limited...

Some people have color blindness

- 8% of males
- 0.5% of females

colors that would be hard for red-green colorblind people to distinguish



Our color vision is limited...

Most common forms of color blindness - red-green called deuteranopia

Google: normal



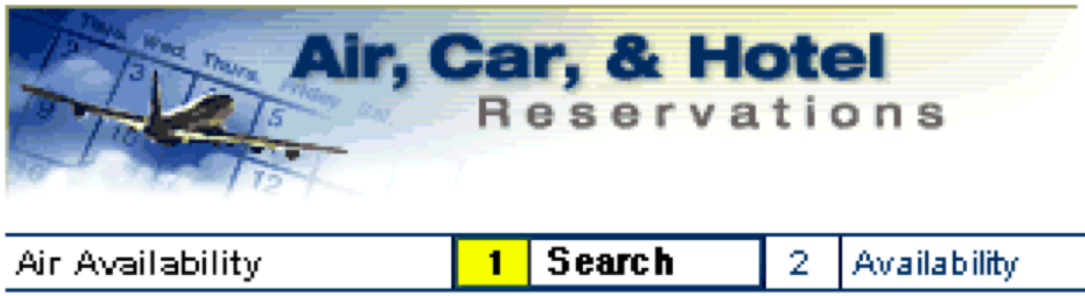
Google: deuteranopia



Our color vision is limited...

Don't use colors only! Also rely on other things like shapes or cues

E.g., Let's improve ITN.net

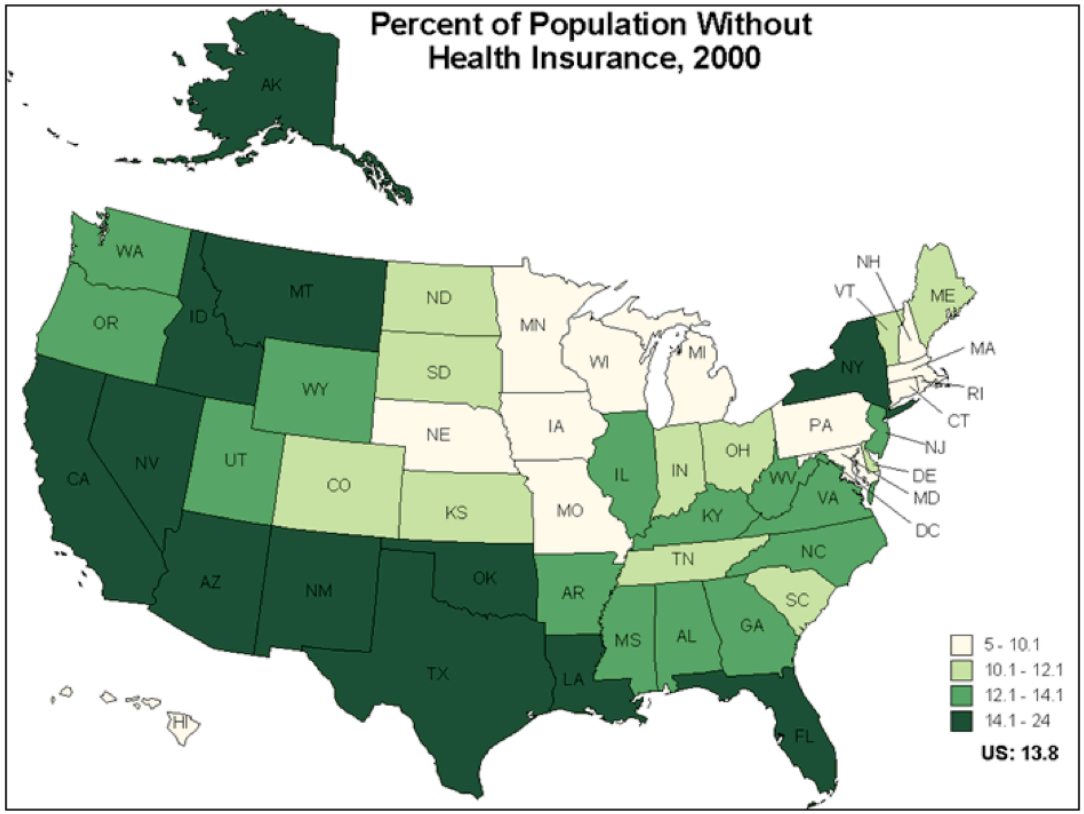


How *they* improved it:



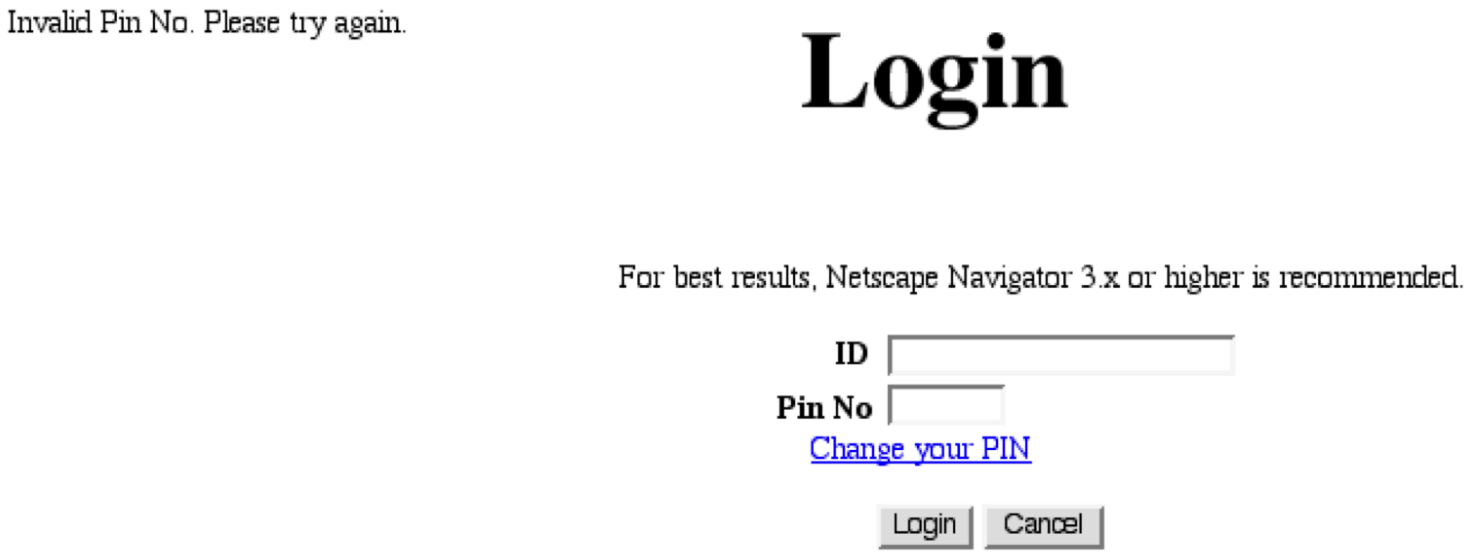
Our color vision is limited...

Don't use subtle color differences. Should still look different in gray scales. Bad examples below....



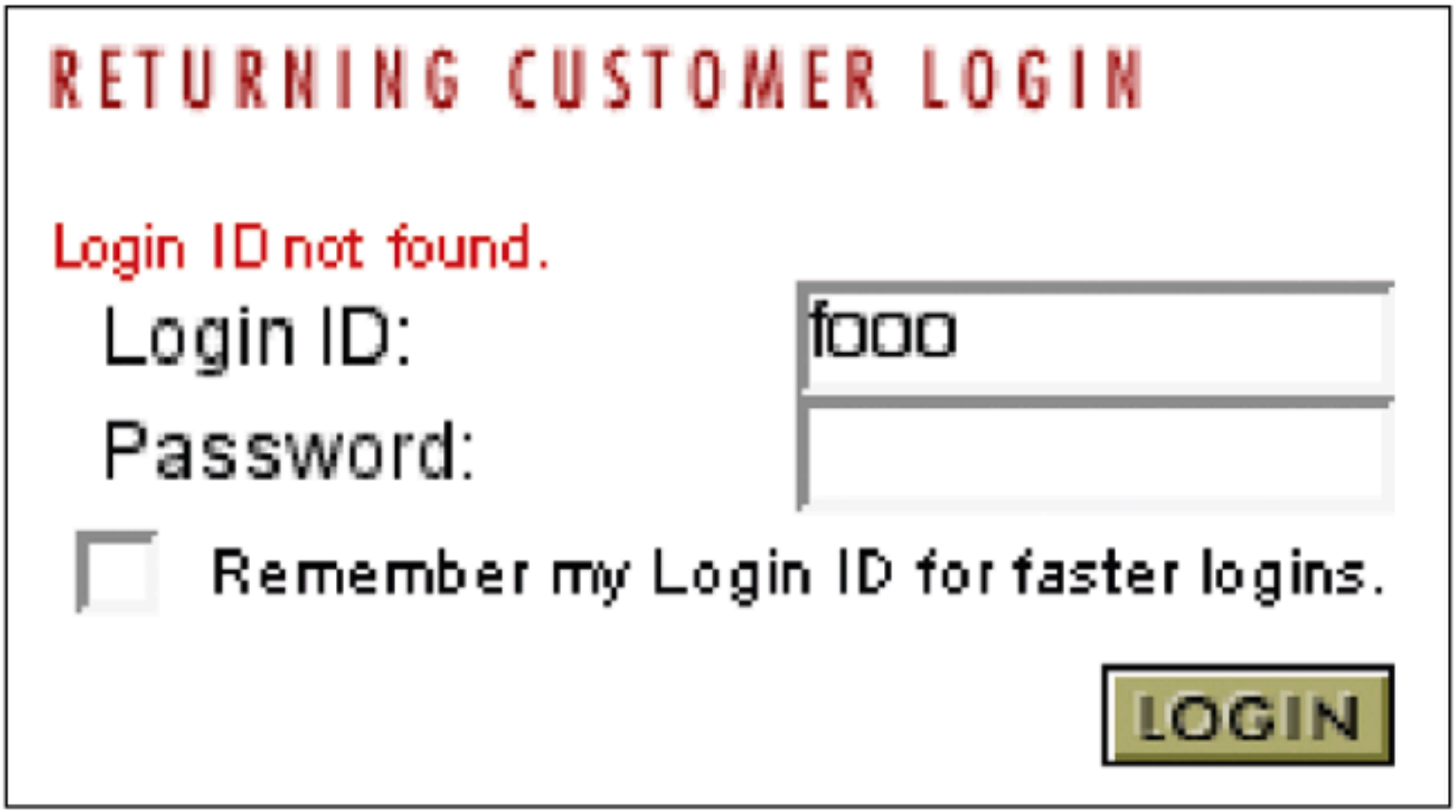
Our peripheral vision is poor...

Our view of vision is actually narrow...



Our peripheral vision is poor...

Bad design...



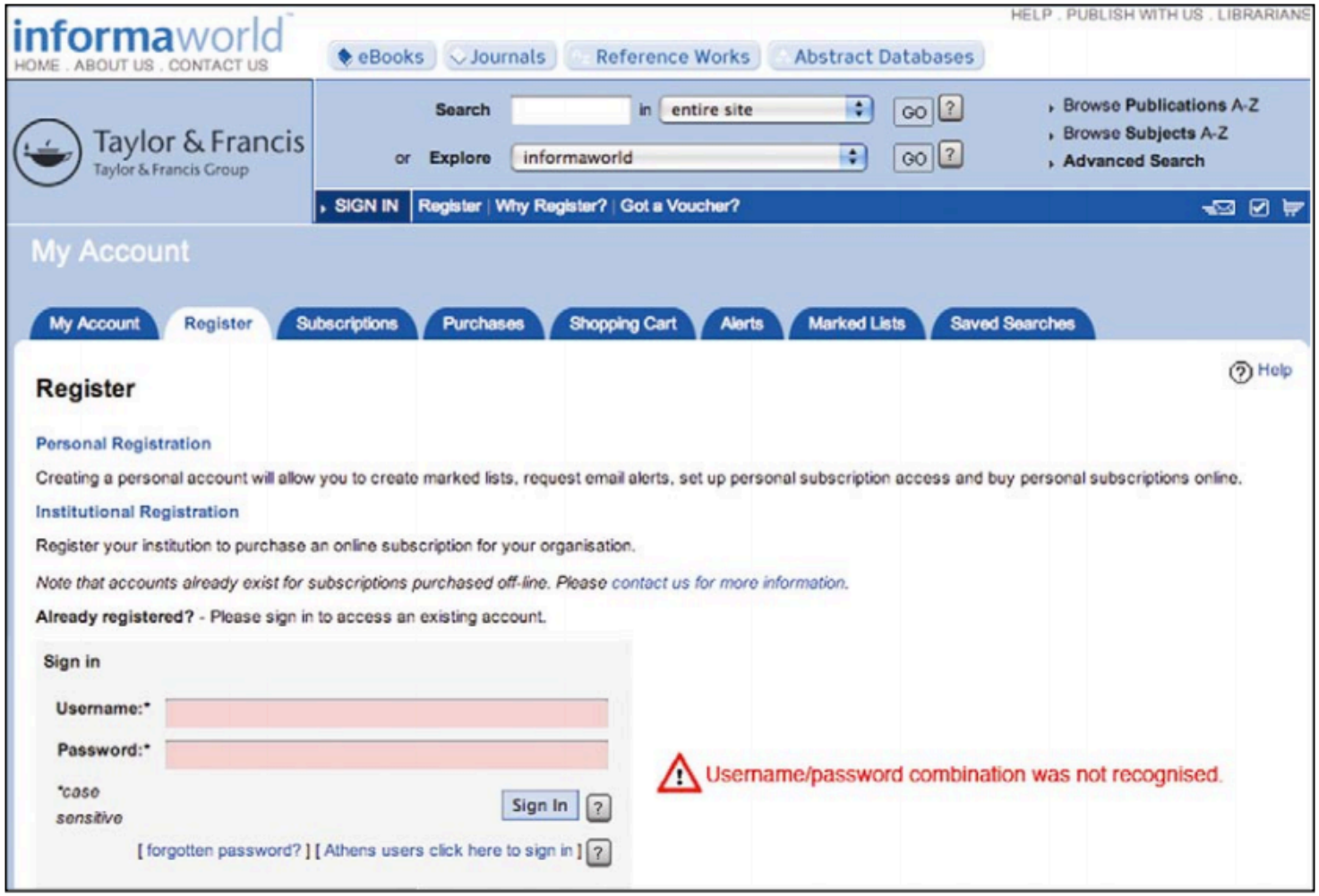
Our peripheral vision is poor...

Simulating the fovea...



Our peripheral vision is poor...

Better design...



Our peripheral vision is poor...

Better design...



HCI and vision

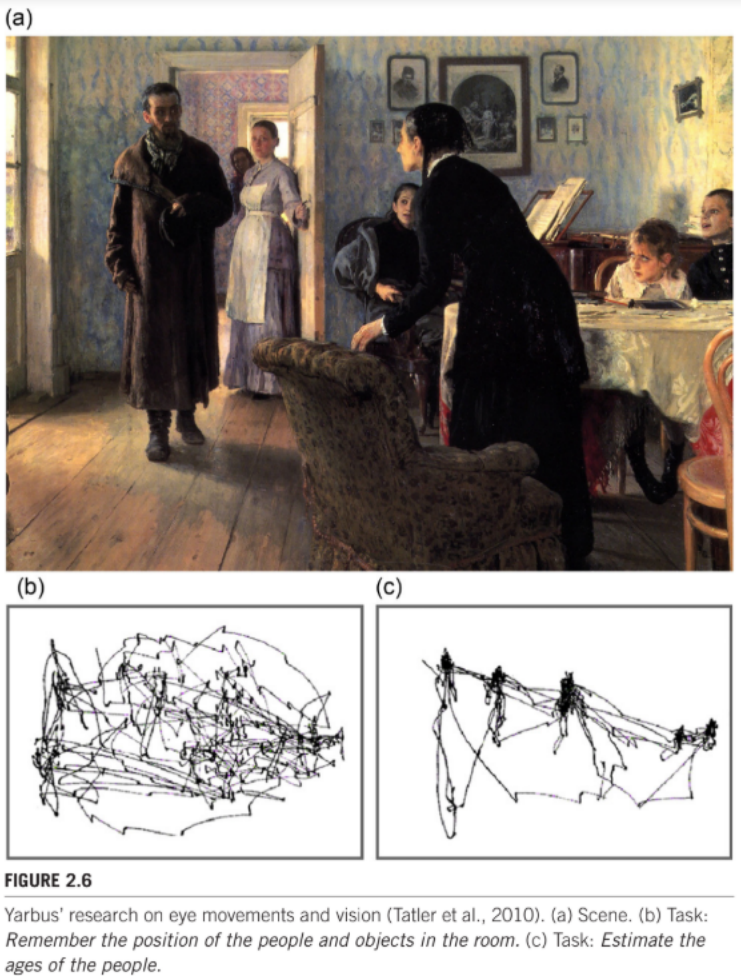


FIGURE 2.6

Yarbus' research on eye movements and vision (Tatler et al., 2010). (a) Scene. (b) Task: *Remember the position of the people and objects in the room.* (c) Task: *Estimate the ages of the people.*

Figure: Source: Fg 2.6 (Mackenzie)

HCI and vision

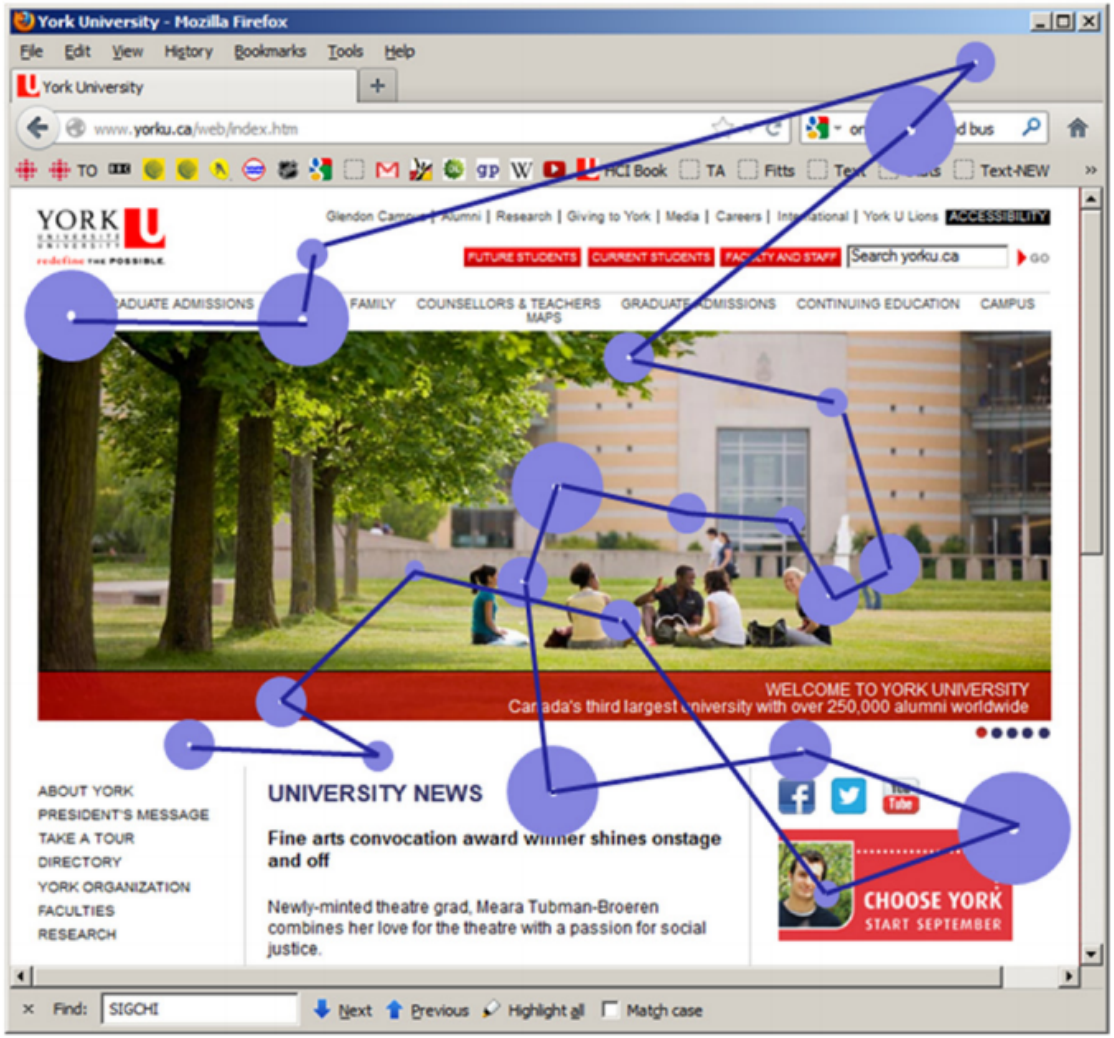


FIGURE 2.7

Scanpath for a user locating content on a web page.

HCI and vision

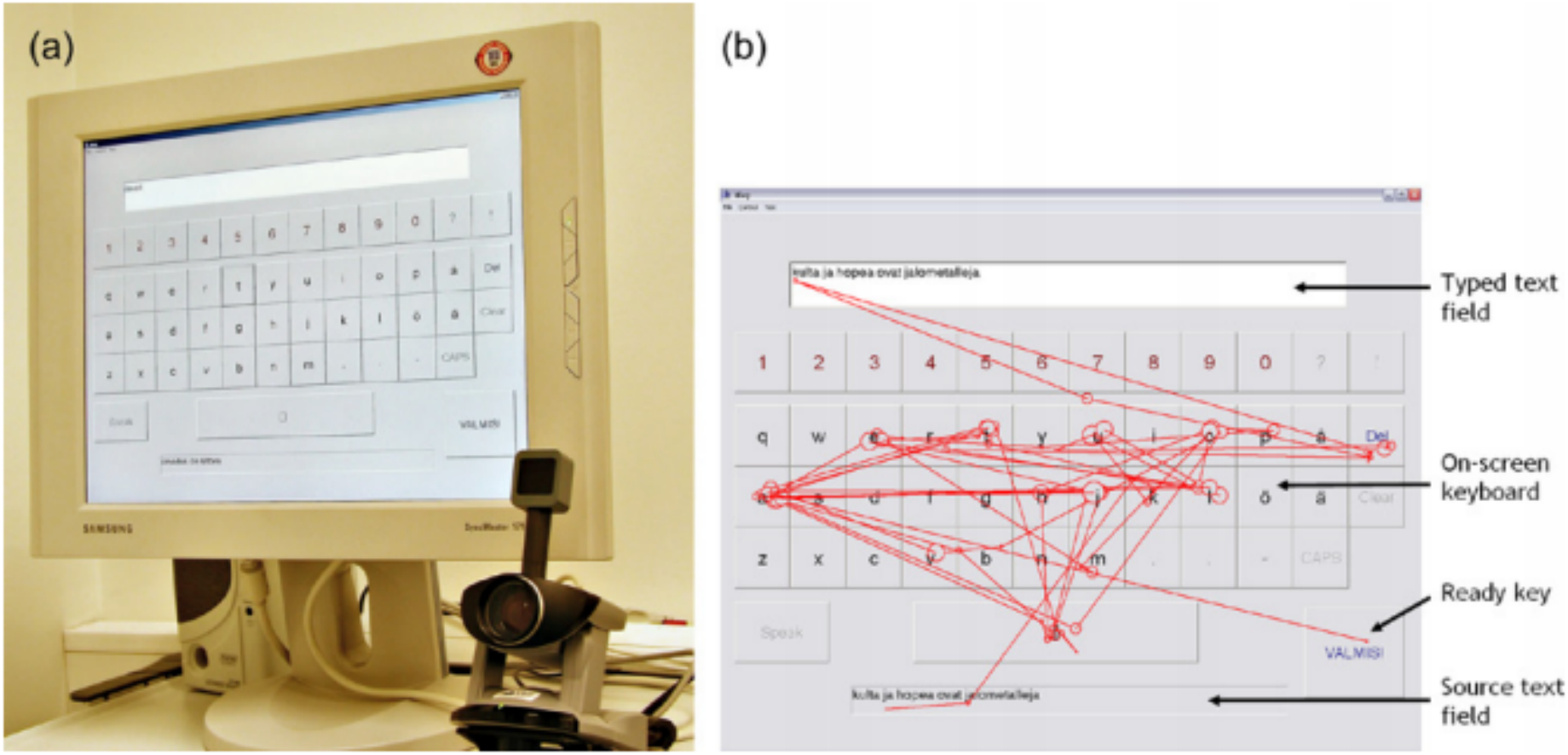


Figure: Source: Fg 2.13 (Mackenzie)

Design implications

- **Don't believe what users say**; instead understand their goals and knowledge (and what they don't know)
- Always exploit **structure** rules
- Use **color** very carefully
- Humans can focus only at **very tiny spot**. Thus put where users are **looking**
- Use **affordance**, **convention**, **constraints**, **mapping** can increase visual search speed

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- Design implications

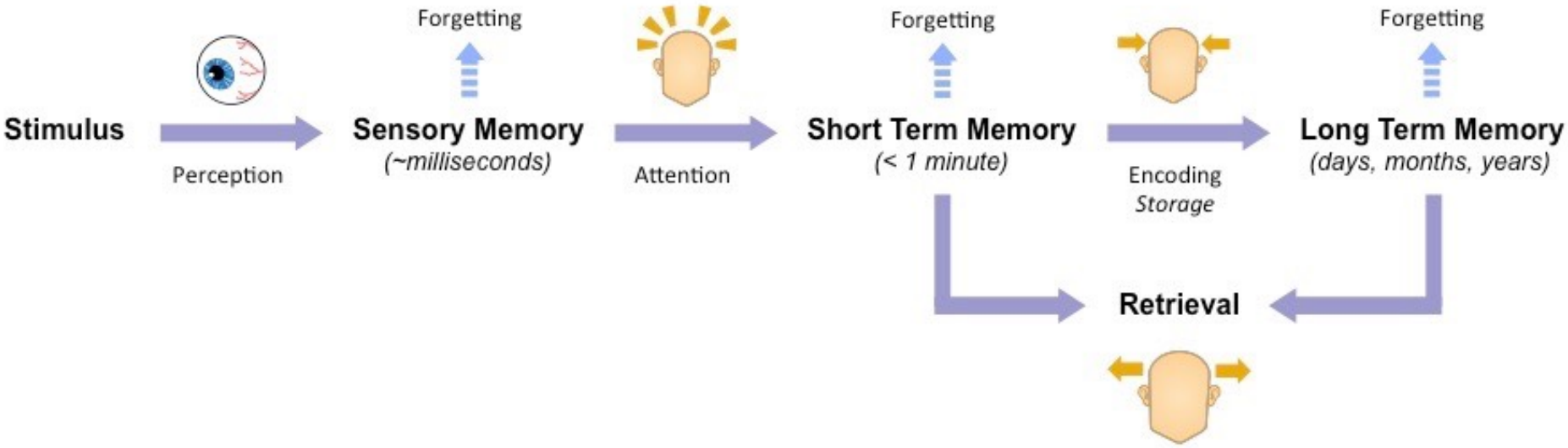
3 Personality

4 Motivation

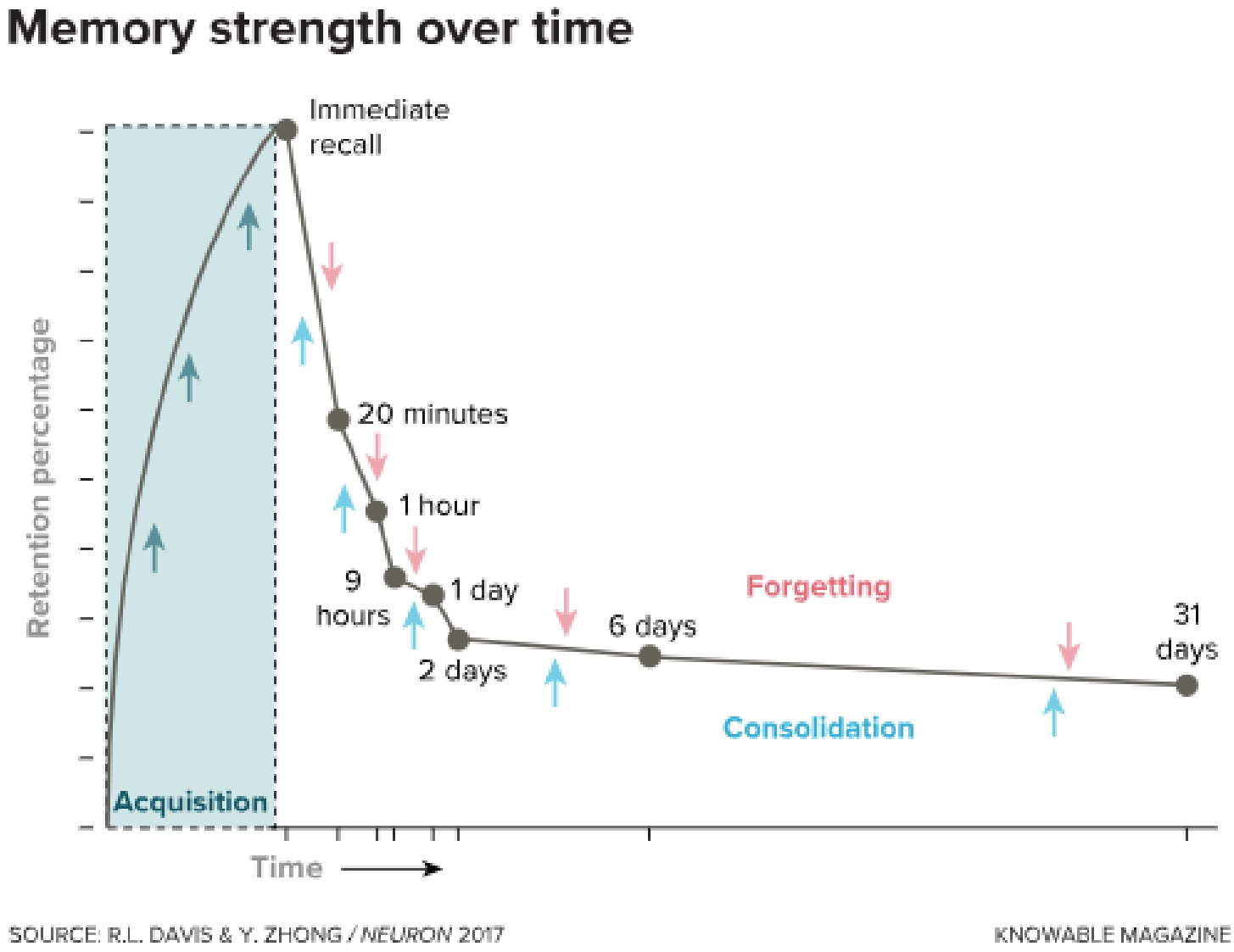
5 Cognition
- Attention
- Learning
- Reasoning
- Design implications

Our memory is imperfect

- Short Term Memory (STM or WM)
- Long Term Memory (LTM)



Our memory is imperfect



Short-term memory

- Also known as working memory
- Can remember around 7 (± 2) unrelated items (3-5 are better estimates, according to Jeff)
- Can stay for around $<$1 minute
- **Chunking** improves our short-term memory

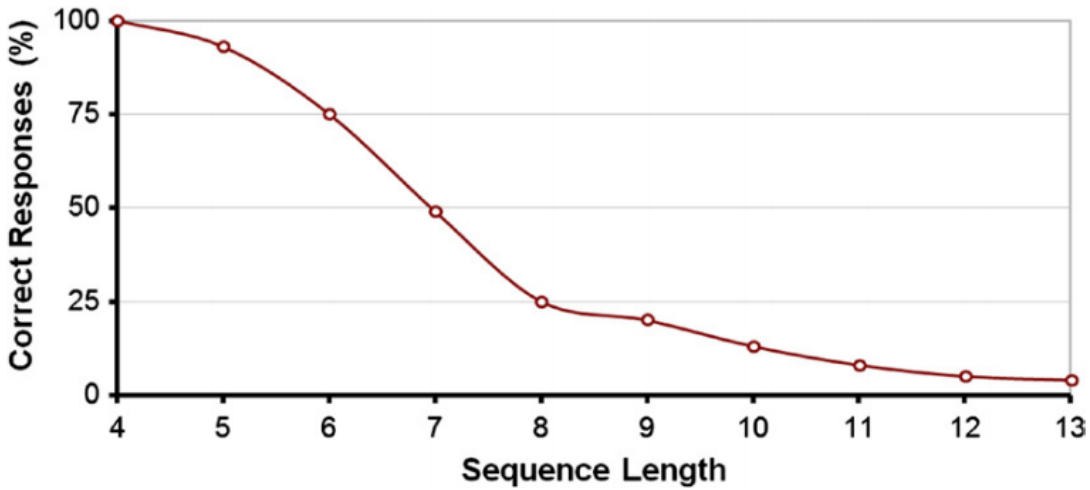
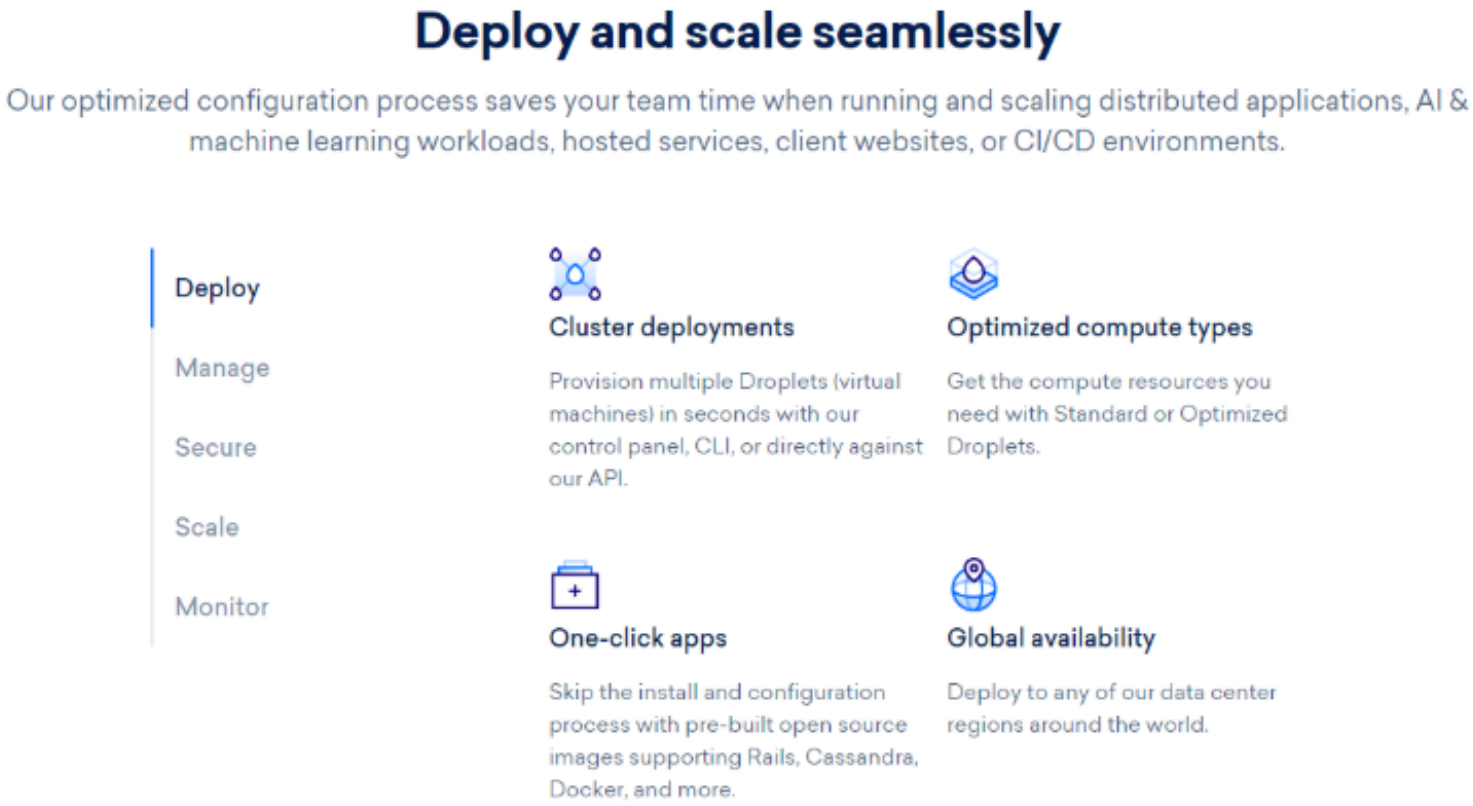


Figure: Source: Figure 2.18 (Mackenzie): Digital lengths and memory

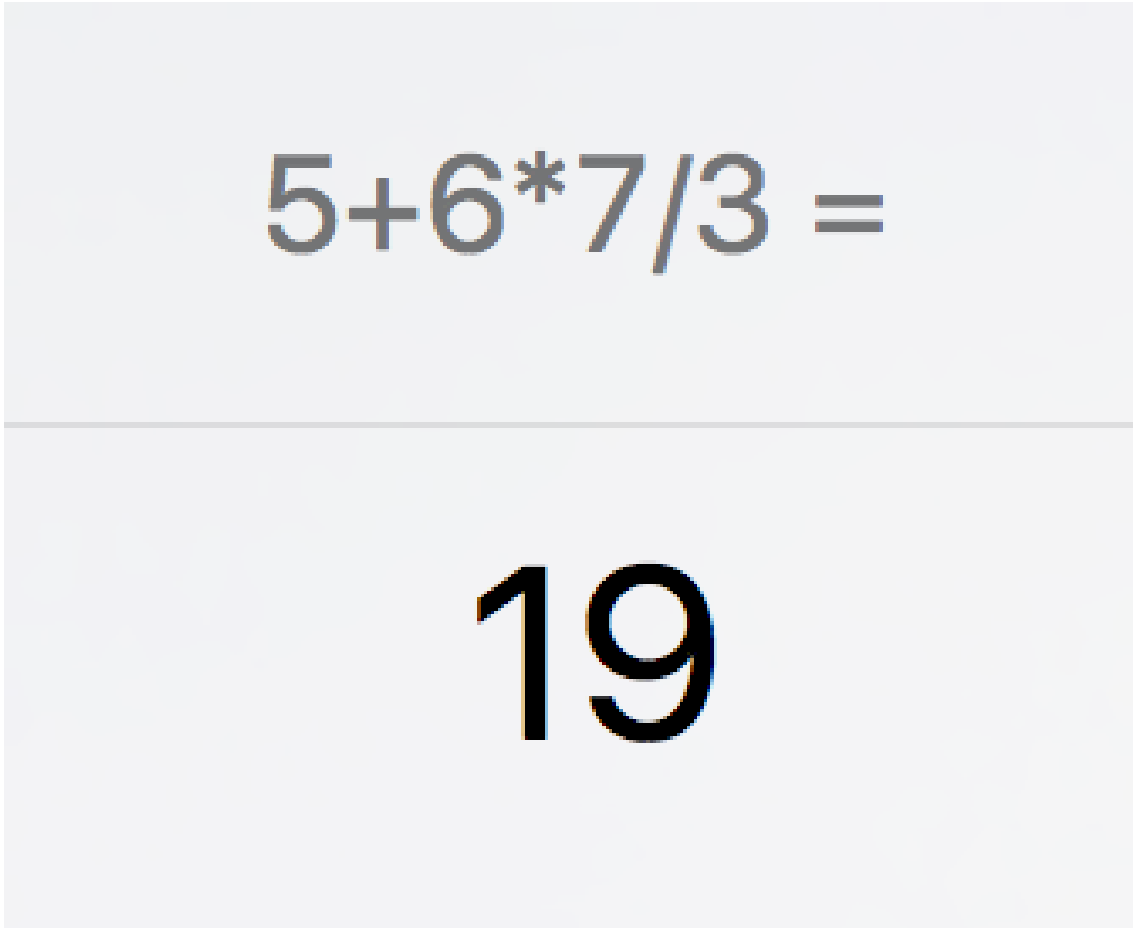
Chunking


Deploy and scale seamlessly
Our optimized configuration process saves your team time when running and scaling distributed applications, AI & machine learning workloads, hosted services, client websites, or CI/CD environments.
Deploy
Manage
Secure
Scale
Monitor
Cluster deployments
Provision multiple Droplets (virtual machines) in seconds with our control panel, CLI, or directly against our API.
Optimized compute types
Get the compute resources you need with Standard or Optimized Droplets.
One-click apps
Skip the install and configuration process with pre-built open source images supporting Rails, Cassandra, Docker, and more.
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Deploy to any of our data center regions around the world.

Long-term memory

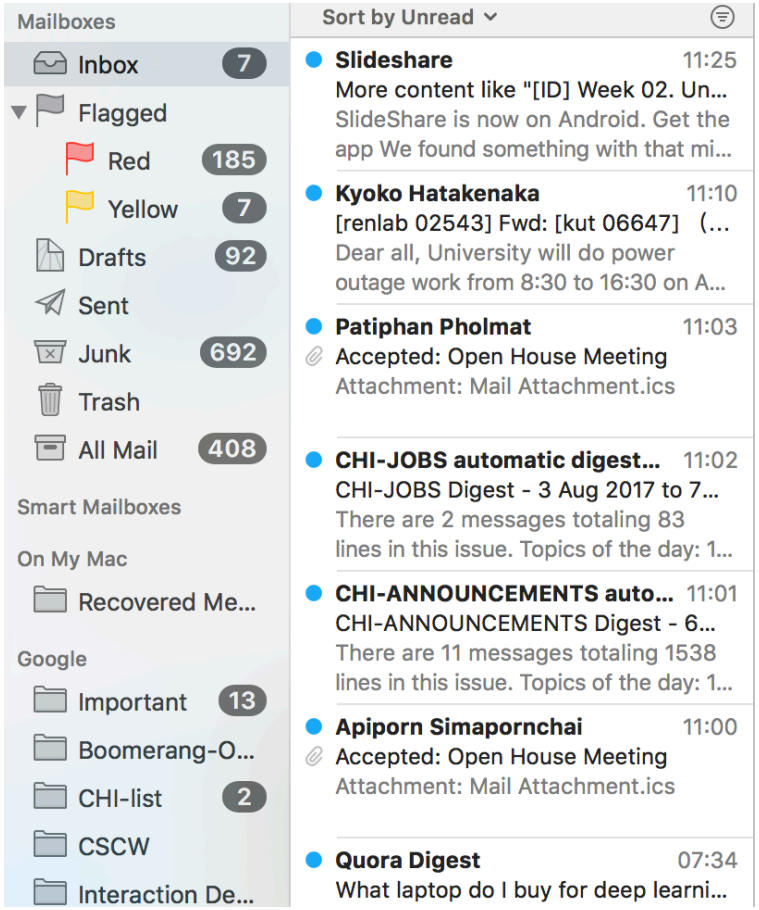
- Stores life time experience but prone to error and biases
- Similar experience trigger same patterns > **recognition**
- Internal neural activity triggers pattern > **recall**
- Why we forget?
  - **Decay theory**: proposes that memory fades due to mere passage of time - active rehearsing information is believed to counter this temporal decline
  - **Interference theory**: proposes that similar information can make memories less accessible

Provide external memory aids



Provide external memory aids

The use of flagging helps...



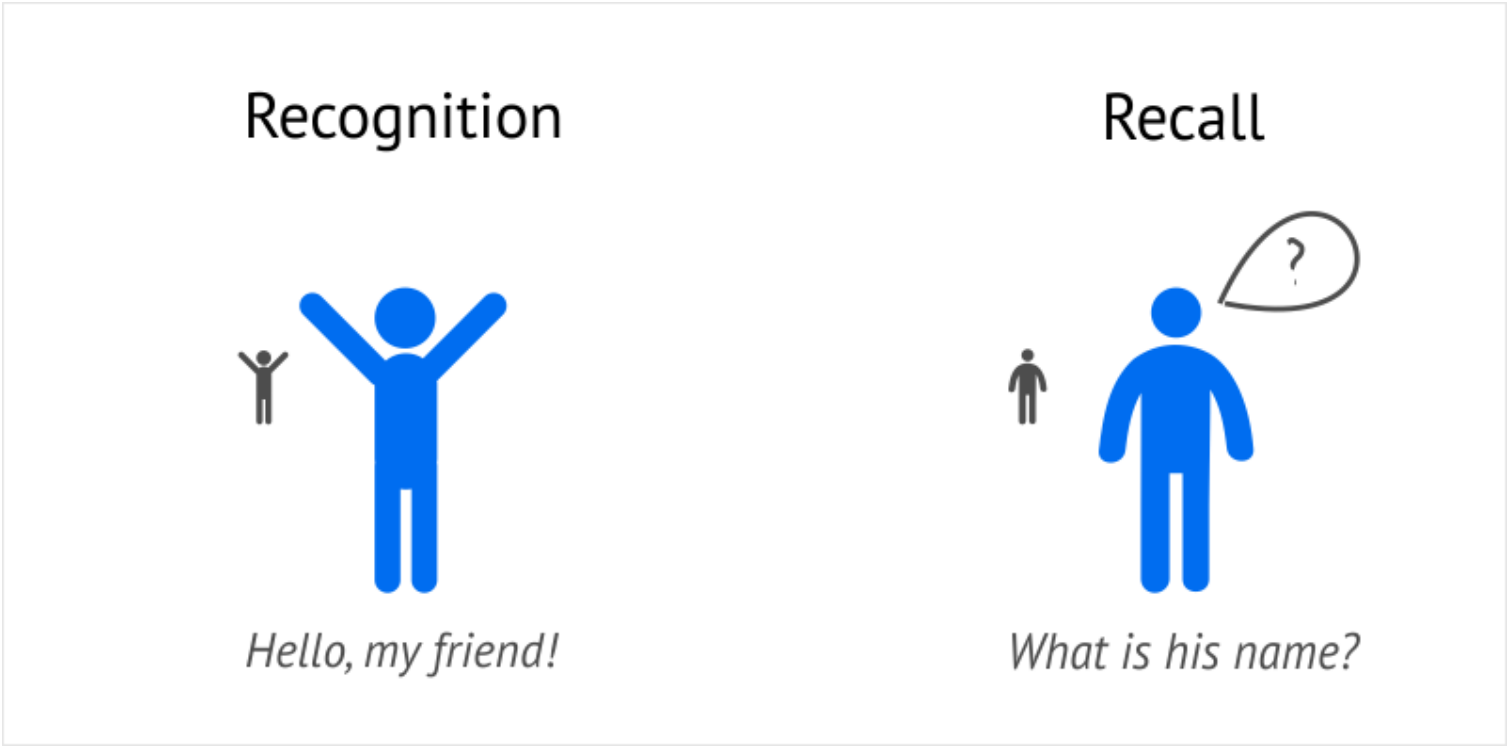
Provide external memory aids

History of commands helps...

```
STIU-M-00101:~ csilpasuwanchai$ history
    1  ipconfig
    2  ifconfig
    3  ping 10.10.101.62
    4  ping Server returned error: "334
    5  Server returned error: "334
    6  ping smtp.office365.com
    7  java -version
    8  cd /System/Library/Java/
    9  ls
   10  ls
   11  cd ..
   12  ls
   13  cd ..
   14  ls
   15  cd /Library/
   16  ls
```

Recognition much faster than recall

- Use menus and metaphors
- Use auto-completion if recall is really needed



Design Implications

- Use **convention** and **consistency**
- Use **recognition** instead of recall when possible (but do not forget ways for novice to transition to experts)
- Use **chunking** when possible; 3-5 rule
- **Put the knowledge in of the world** (Norman, 1988), e.g., bookmarks, history of commands, tagging, time stamping, reminders, marked emails, pwd etc.

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Personality: Same app, different preferences

Two people use the same fitness app. They want different experiences.

"Let me compete."

- Share a run with friends.
- Join a group challenge.
- Compare progress with others.

"Let me focus privately."

- Keep activity private.
- Set a personal goal.
- Compare with yesterday's self.

Ask: Which would you prefer? Would your answer change with the situation?

Personality describes broad tendencies. Goals, experience, and context also matter.

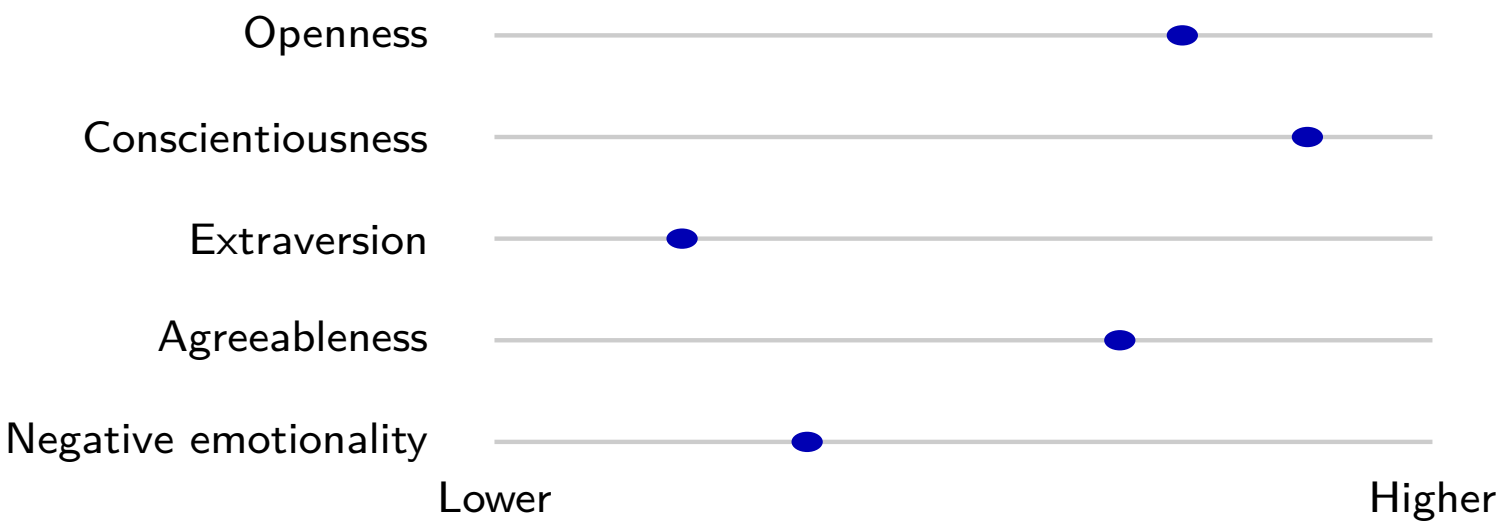
The Big Five: Five dimensions, not five types

Dimension	One end	Other end
Openness	Prefer familiar approaches	Enjoy new ideas
Conscientiousness	More spontaneous	More organized
Extraversion	More reserved	More outgoing
Agreeableness	More skeptical, competitive	More trusting, cooperative
Negative emotionality	More emotionally steady	More prone to worry

People fall along each dimension. Neither end is a diagnosis or a measure of worth.

Soto and John (2017), Big Five Inventory–2. Descriptions simplified for teaching.

A profile is more useful than a label



This fictional person is reserved and organized, and enjoys new ideas.

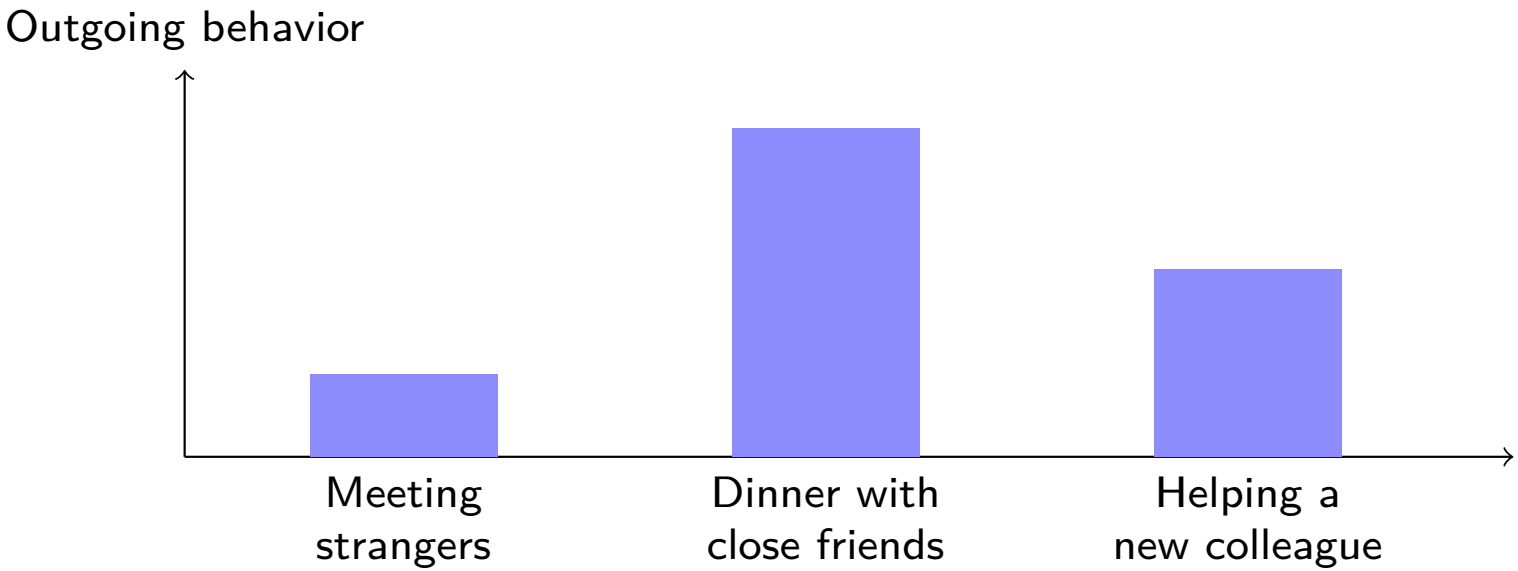
A reserved person may dislike public leaderboards but enjoy sharing privately with close friends."

Ask directly: "Would you like to share this activity? With whom?

Illustrative profile, not measured scores. Trait descriptions do not prescribe interface choices.

Surprise: One person can act very differently

Being reserved on average does not mean being quiet in every situation.



Studies tracking people through daily life found substantial variation within the same person.

Remember

Personality is a tendency, not a script. Design for the person **and the situation**.

Fleeson (2001). Bar heights illustrate a fictional person; they are not study data.

Personality: Design implications

- **Offer meaningful choices:** private goals or optional group challenges.
- **Let users control the experience:** sharing, reminders, and public visibility.
- **Ask preferences directly:** do not infer a personality from a few clicks.
- **Test with different people:** a feature you enjoy may frustrate someone else.

Exercise

A fitness app automatically publishes every workout. Propose one change for someone who wants privacy and one for someone who enjoys sharing.

Possible answer: private by default, with optional sharing and group challenges.

1. Perception
   - Biases
   - Structure
   - Limitations
   - HCI and vision
   - Design implications
2. Memory
   - Design implications
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   - Attention
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   - Reasoning
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Motivation: Why would someone keep using it?

Reason	Everyday example	What drives it?
Enjoyment	"I play guitar because it is fun."	The activity itself
Personal value	"I exercise to stay healthy."	A valued outcome
External reward	"I walk to earn shopping points."	A separate reward

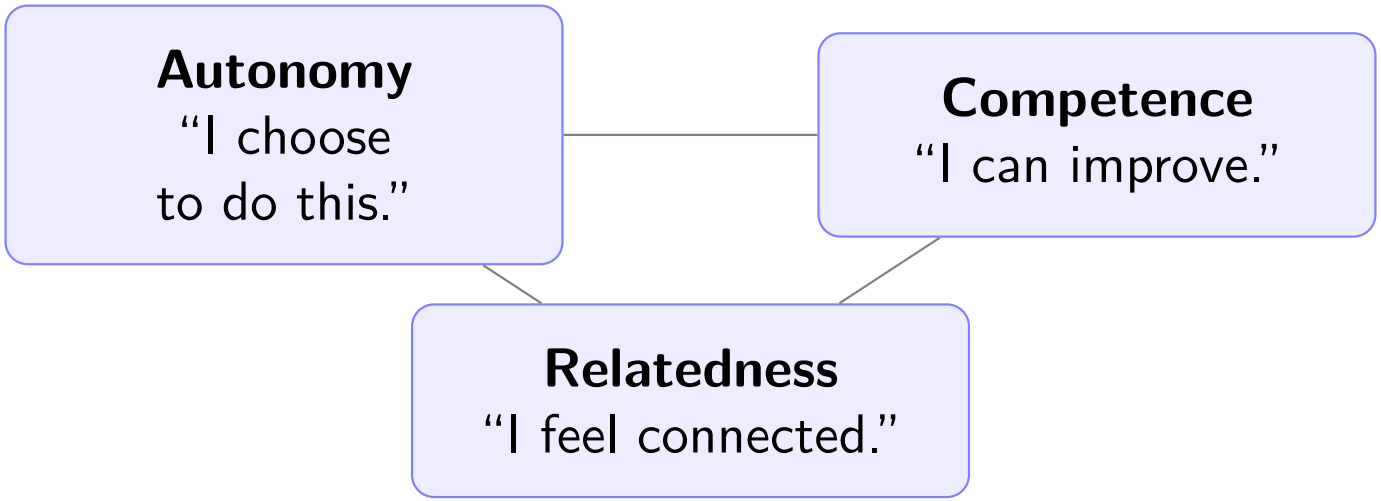
Intrinsic motivation: doing an activity for its own enjoyment.

Extrinsic motivation: doing it for a separate outcome. This can be personally valued, not just imposed.

Ryan and Deci (2000). People can have several motives at once.

Three needs that support motivation

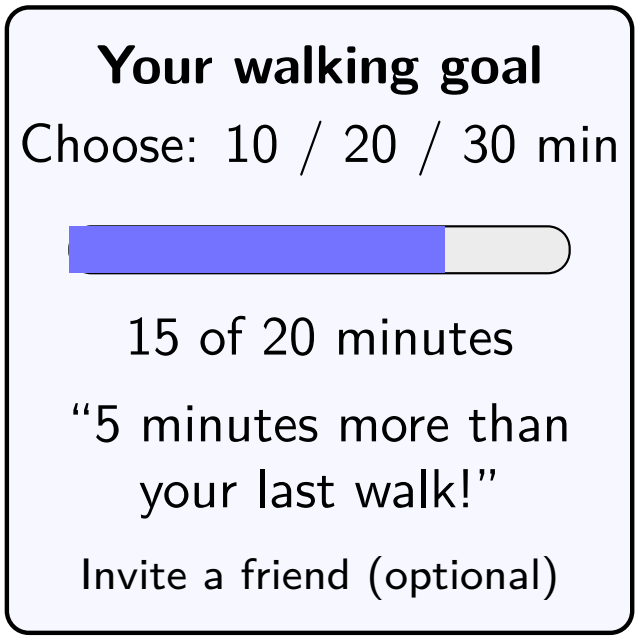
Self-Determination Theory: support autonomy, competence, and relatedness.



Ask: Which need does your favorite game or hobby support?

Ryan and Deci (2000). Relatedness can come from support; it does not require public competition.

Example: Make progress visible



- **Autonomy:** choose a realistic goal.
- **Competence:** see specific progress and useful feedback.
- **Relatedness:** invite a supportive friend if wanted.

Ask: What happens if the app forces a goal and shames you for missing it?

Illustrative design applying Self-Determination Theory; not an evaluated intervention.

Surprise: A reward can reduce interest

Predict: If someone enjoys drawing, will paying them for every picture always make them want to draw more?

Before	**After rewards become expected**
"I draw because I enjoy it."	"What do I get for drawing?"

A review of 128 experiments found that expected tangible rewards could reduce later voluntary engagement in already interesting activities.

This does **not** mean all rewards are harmful. Reward type, feedback, and context matter.

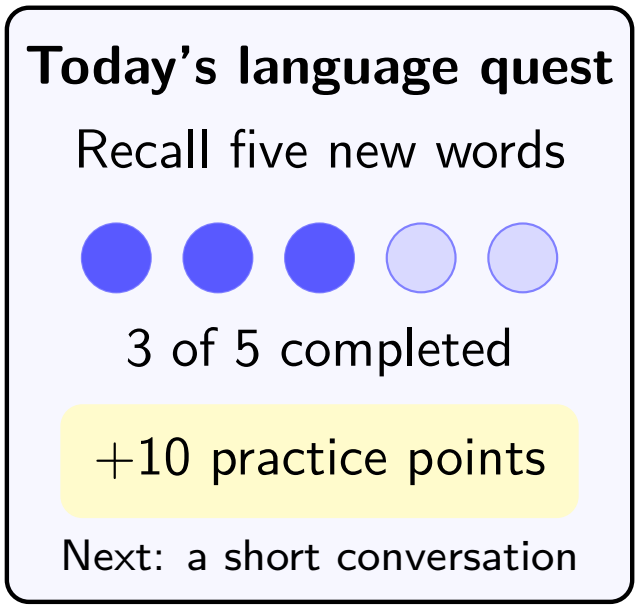
Remember

More rewards do not automatically mean more motivation. Build interest and competence, not only points.

Deci, Koestner, and Ryan (1999). Dialogue illustrates the mechanism, not a study quotation.

Gamification: Borrow ideas from games

Gamification uses game-design elements in activities that are not games.



- **Quest:** a clear, achievable challenge.
- **Progress:** see how far you have come.
- **Points or badges:** mark achievements.
- **Levels:** introduce harder tasks as skills grow.

Ask: Are users learning words, or only collecting points?

Deterding et al. (2011). Interface is an illustrative example.

Why might game elements motivate?

Start with the psychological need, then choose a feature.

Need	Possible feature	What users experience
Autonomy	Choose a quest or difficulty	"I decide how to participate."
Competence	Levels and useful progress feedback	"I can see myself improving."
Relatedness	Optional team challenges	"We support each other."

Remember

Points do not automatically create motivation. The experience behind the feature matters.

Ryan and Deci (2000); Sailer et al. (2017). These are design possibilities, not guaranteed effects.

The same feature can encourage or pressure

Feature	Can support	Can create problems	
Streak	A regular practice routine	Fear of losing progress after a missed day	**Design**
Leaderboard	Friendly, optional competition	Discouragement for people near the bottom	
Points	Feedback on meaningful practice	Repeating easy actions just to earn points	

implications: allow rest and recovery, make competition optional, and reward meaningful progress.

Test the real goal: compare later word recall, enjoyment, and pressure—not only clicks or daily visits.

Illustrative benefits and risks to investigate; Sailer et al. (2017) shows effects differ across elements.

Motivation: Design implications

- **Support choice:** let users set goals and adjust reminders.
- **Support improvement:** give achievable steps and specific feedback.
- **Support connection:** offer helpful social features without forcing comparison.
- **Look beyond points:** measure whether users achieve goals they value.

Exercise

A language app says: "Miss today and lose your 100-day streak!" Rewrite it to offer a manageable next step without guilt.

Possible answer: "Have two minutes? Practice five words, or choose a better time."

Design suggestions informed by Ryan and Deci (2000); effects depend on context.

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Attention

- Our attention is limited
- Goldfish has an attention span of 9s, guess how much is of humans? 8s! (In 2000, our attention span is 12s)
- Do you remember what I discuss in the previous slide? What did you eat yesterday morning?



Our attention is on the goal, not the tools/person

https://www.youtube.com/watch?v=FWSxSQsspiQ

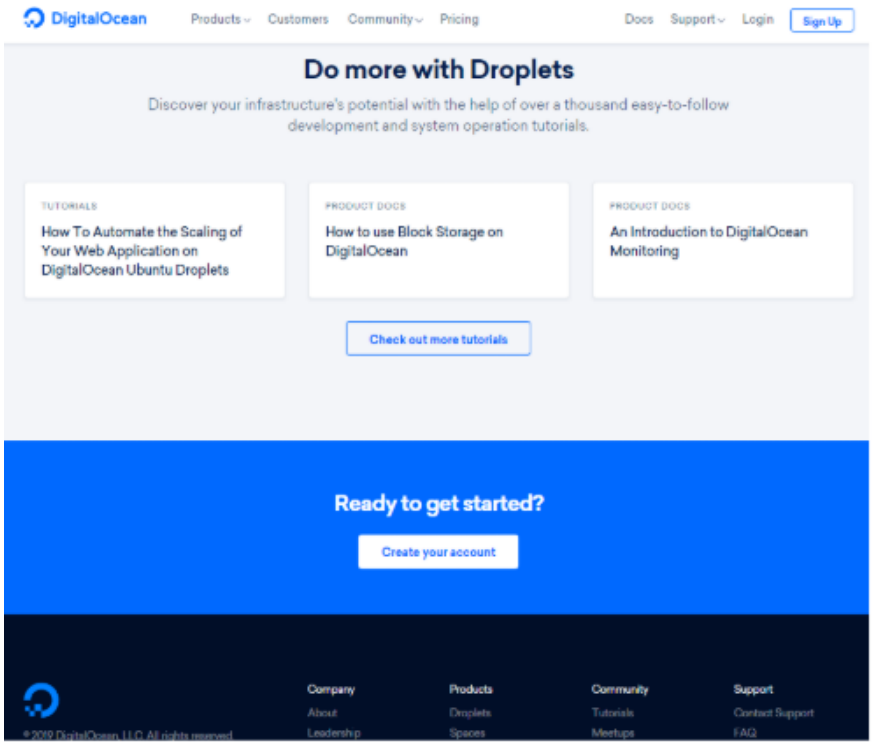


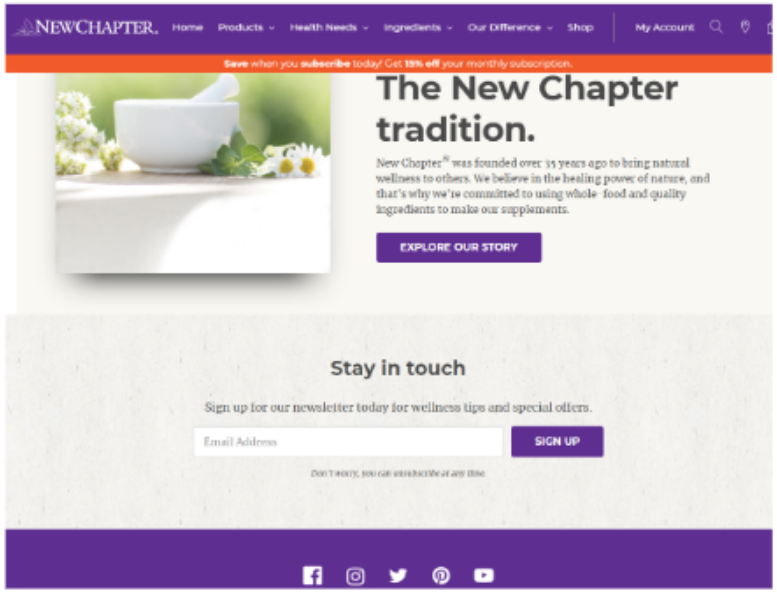
Loose ends

- When we finish our goal, we often forget the "loose ends" of tasks
  - Turning headlights of car off
  - Forgetting to take your ATM card after withdrawing money



End with action

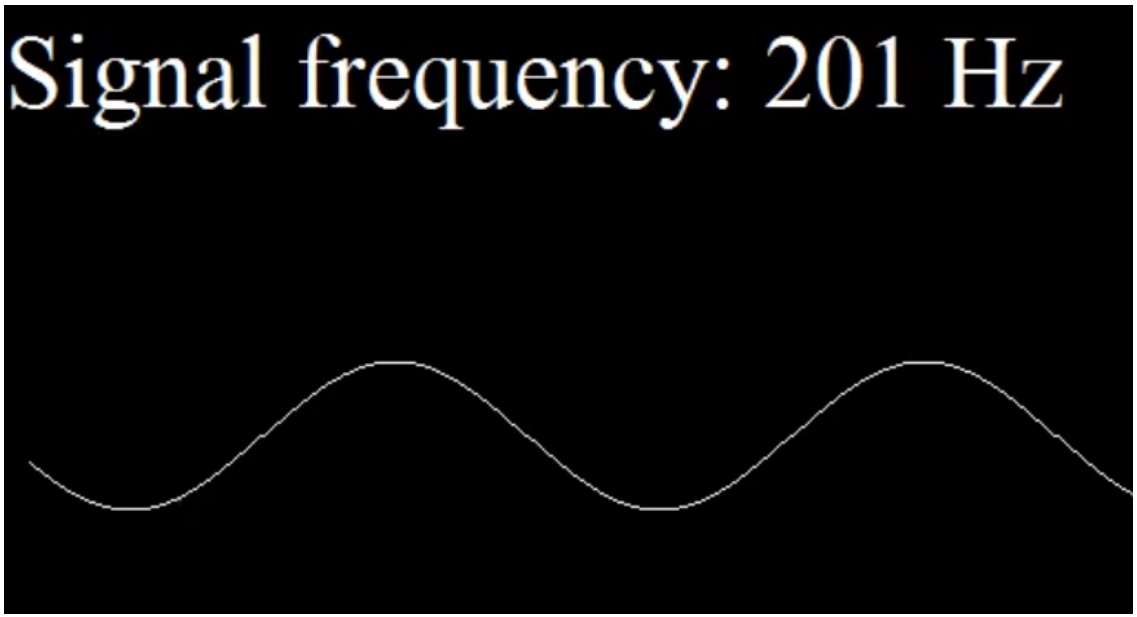




Hearing capabilities

- Human hearing begins with sounds of 0-10dB. Conversational speech is about 50-70 dB. Painful sound is about 120-140 dB
- Pitch is the frequency and human can perceive sounds in the range of 20Hz to 20,000Hz (mine was 200Hz! How about yours?)

`https://www.youtube.com/watch?v=qNf9nzvnd1k`



Multitasking

- Humans cannot really multi-task, we can only quickly switch between multiple tasks in ms

  `https://www.youtube.com/watch?v=vJG698U2Mvo`

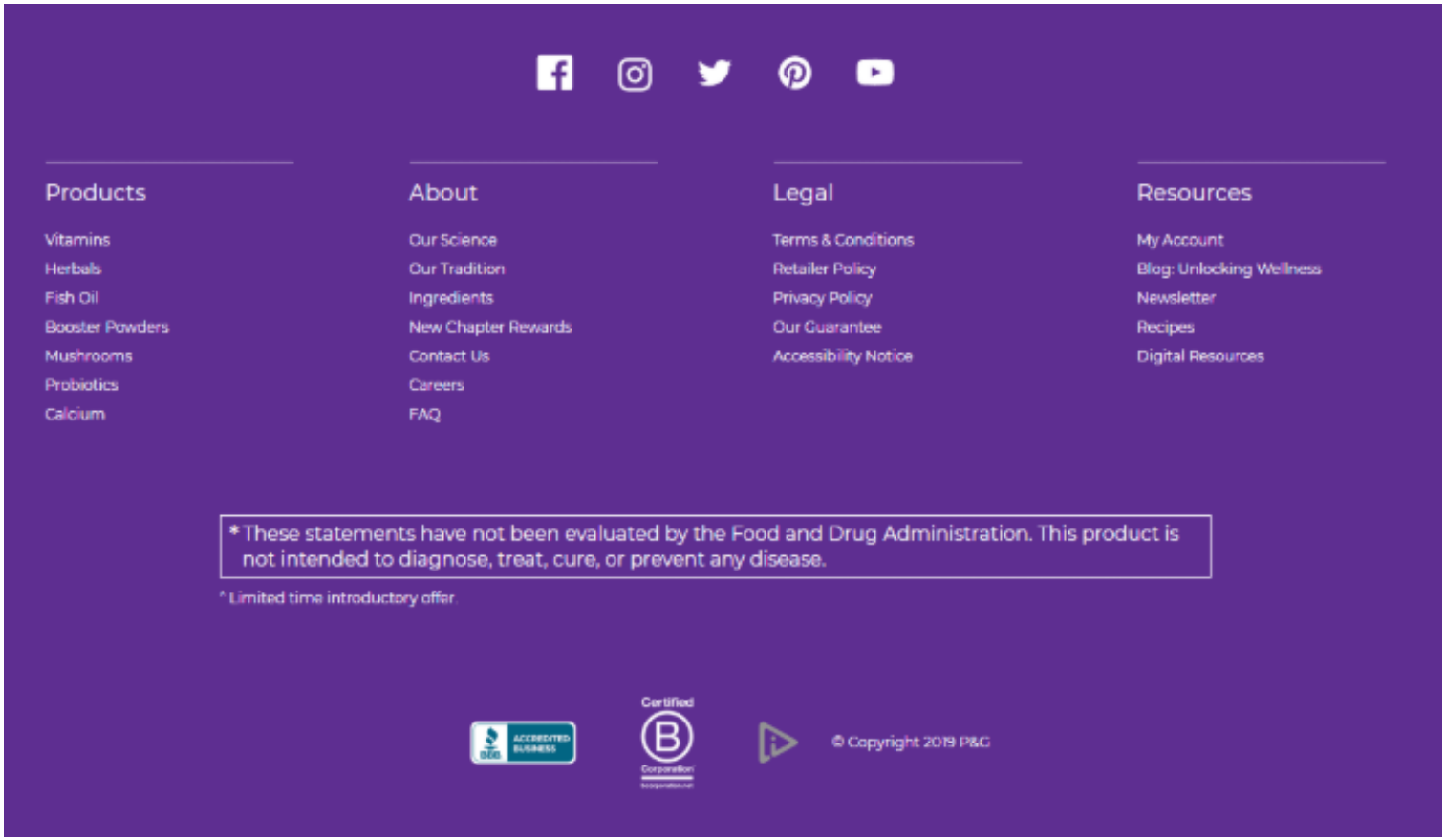


Learning

- **Convention** helps learning
- **Feedback** helps learning
- **Risk** is low
- **Reward** is high

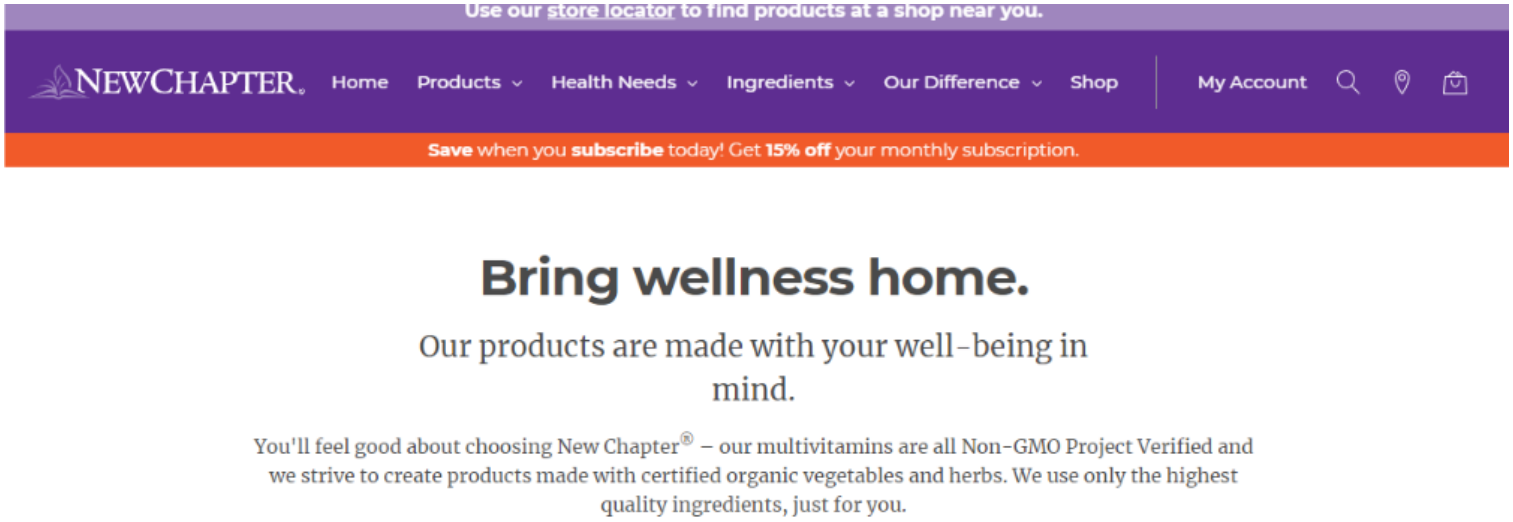
Convention helps learning

Always preach...follows convention!



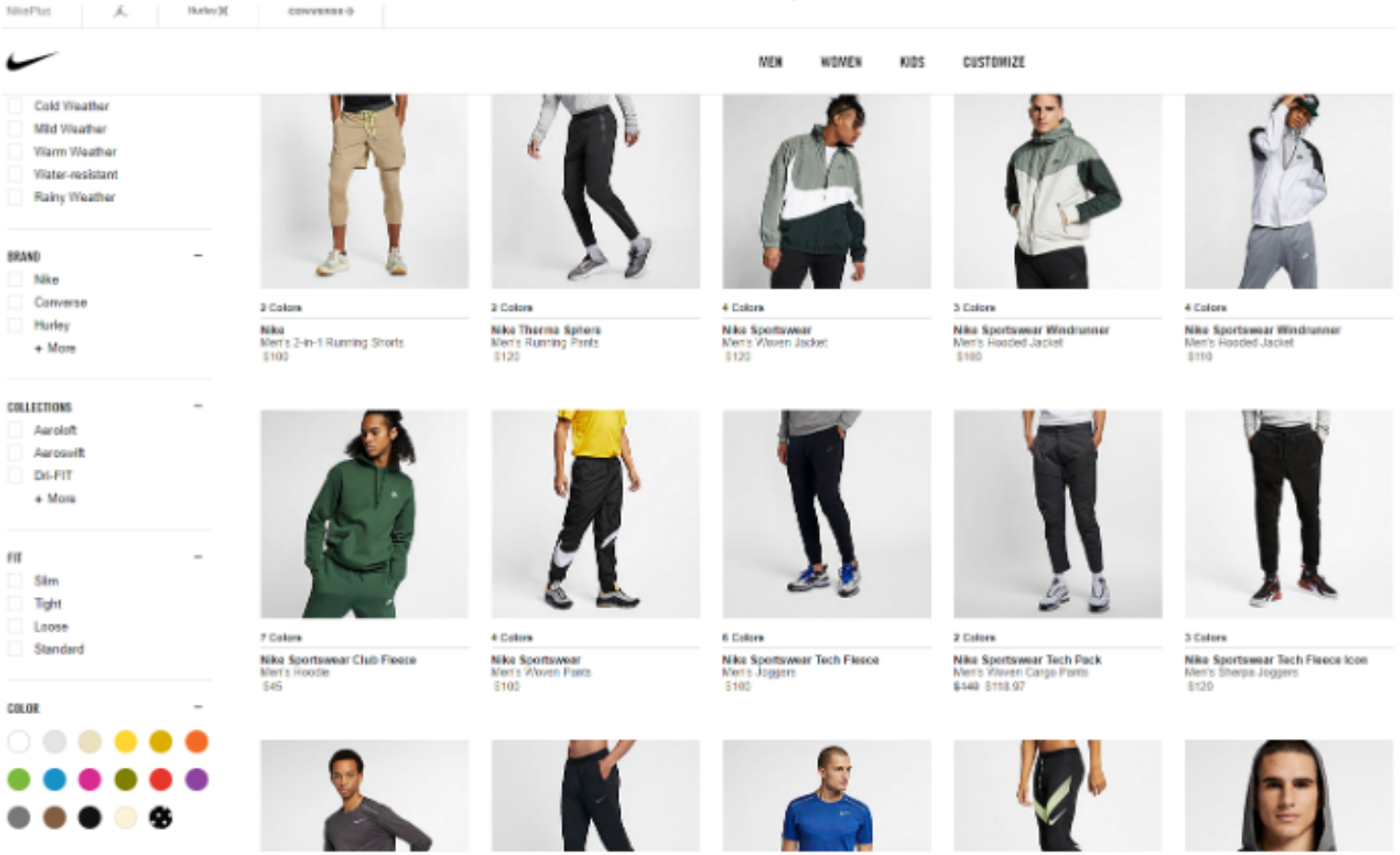
Convention helps learning

Always preach...follows convention!



Convention helps learning

Convention is about "stealing" but make it better



Feedback helps learning

Feedback informs us how to become better



Risk is low

- **Cheap failures** - no ways to make errors; easy to recover



Rewards

- If you decide to break the cycle and reinvent something new
  - make sure its **reward exceeds learning effort**
  - make sure it's **learnable** and able to master over time!
  - even better, **expert can still learn** something, i.e., novice vs. expert mode
- **Games** are fun, because everytime we play, **we get better, to no limits.** (Self Determination Theory, 1985)



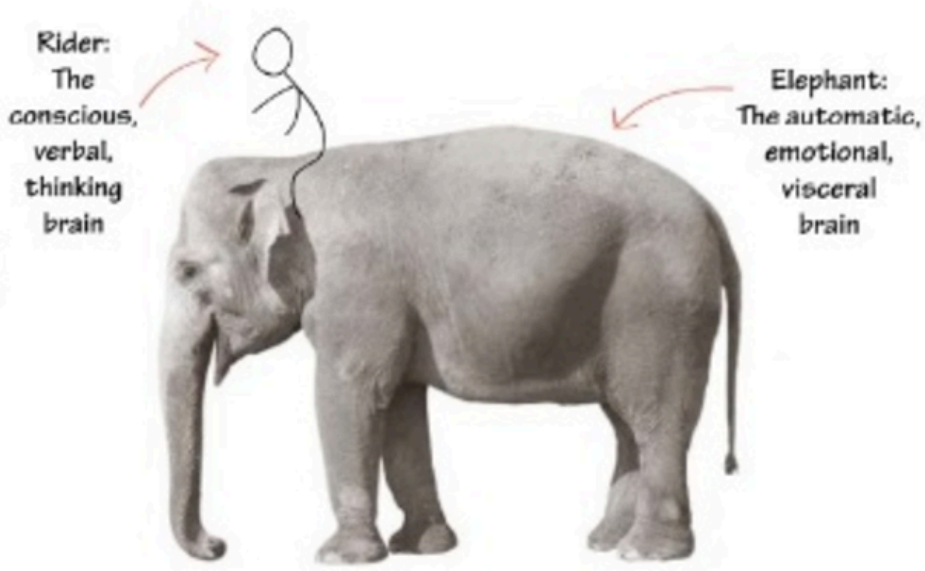
System 1 and System 2

- Our brain has two systems: system 1 and system 2
- **System 1** is the *irrational* brain - fast, automatic, unconscious, yet govern most of our behavior
- **System 2** is the *rational* brain - slow, precise, conscious, "believes" it governs our behaviors

System 1 and 2

One brain, two minds

- Elephant = impulsive mind
- Rider = Rational mind
- Path = the environment



Willpower is the strength of the rider

System 1 and System 2

- A baseball and a bat together cost 110. The bat costs 100 more than the ball. How much does the ball cost?
- System 1 instant answer: 10 (wrong)
- System 2 may reject that answer. Or not.
- System 2 can calculate correct answer; System 1 cannot.

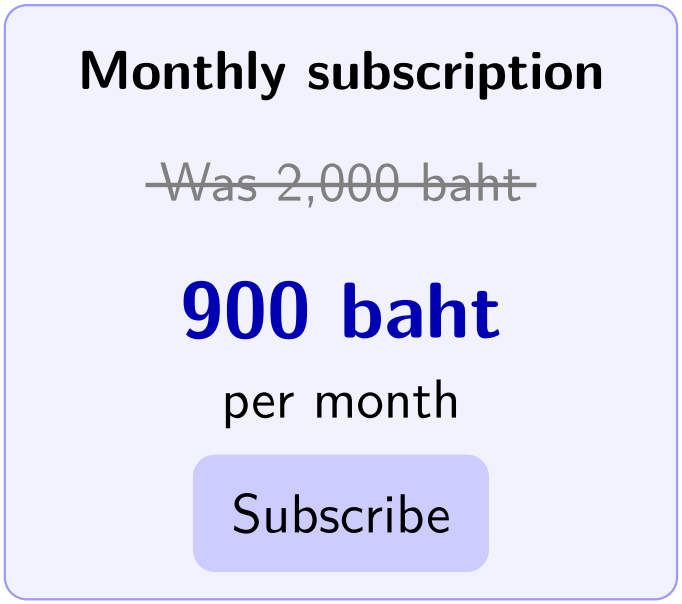
Human decisions are rarely rational

System 1 usually controls decisions but is very biased.

- **Losses** mean more than **gains**
- **Recent** history and strong memories "feel" more
- **Experience** and **intuition** means more than mountains of statistics and data
- People **avoid risks** for potential gains, but **take risks** for potential losses
- Influence by **word** (75% survival rate vs. 25% mortality rate)

1. Anchoring bias

The first number we see can influence our judgment.



Ask: Is 900 baht good value, or does it just look cheap compared with 2,000?

- The original price becomes an **anchor**.
- We may give it too much weight when judging value.

Design implication

Help users compare features, total costs, and alternatives.

Illustrative interface. Tversky and Kahneman (1974).

2. Confirmation bias

We tend to favor evidence that supports what we already believe.

"Brand A makes the best phones."

Phone reviews

"Great camera!"

"Battery runs out quickly."

You already love Brand A.

You trust the positive review, but dismiss the negative one:

"They probably used it wrong."

Ask: Would you judge the same reviews differently for another brand?

Design implication

Help users compare the same criteria across products. In user testing, actively seek evidence against your own design.

Illustrative scenario. Nickerson (1998).

3. Availability bias

Events that are easy to remember can seem more common than they are.

BREAKING NEWS
Dramatic footage
of a plane accident

↓

"Flying feels dangerous."

You see repeated footage of a plane accident.

Before your next flight, accidents suddenly feel very likely—without checking how often they occur.

Ask: Is the risk high, or is the image easy to recall?

Design implication

When showing risks, provide frequencies and context alongside vivid stories or alerts.

Illustrative scenario. Tversky and Kahneman (1974).

4. Loss aversion

A loss can feel more significant than an equivalent gain, depending on the context.

Find money
+500 baht

Lose money
−500 baht

Imagine finding 500 baht in an old jacket.

Now imagine losing 500 baht from your wallet.

Ask: Which feeling would be stronger?

The loss may hurt more than the equal gain feels good.

Design implication

Make changes reversible: offer undo, backups, and previews so users can explore without risking their work.

Illustrative scenario. Kahneman, Knetsch, and Thaler (1991).

5. Sunk-cost fallacy

Past spending can push us to continue something that is no longer worthwhile.



You stay because "I already paid for it."

The ticket money is gone whether you stay or leave.

Ask: Is watching the rest worth another hour of your time?

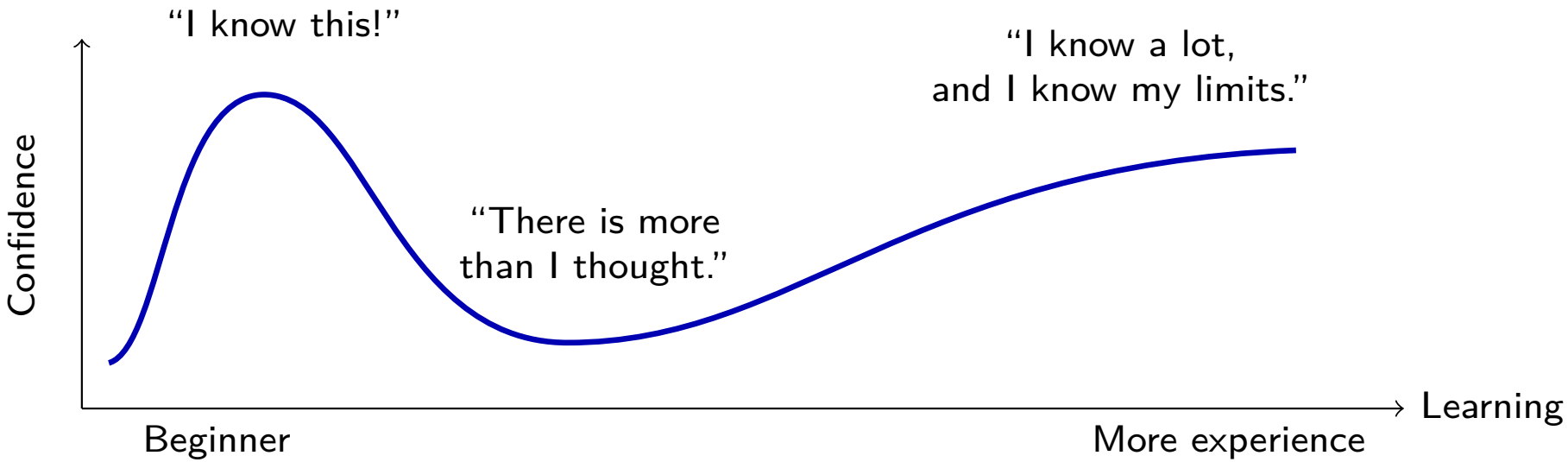
Design implication

When reviewing a design, judge future benefits and costs. Months already spent building it are not a reason to keep a poor solution.

Arkes and Blumer (1985).

6. Dunning–Kruger effect

Limited skill in a task can make it difficult to recognize your own mistakes.



Popular learning-journey sketch, not measured data or the original Dunning–Kruger graph.

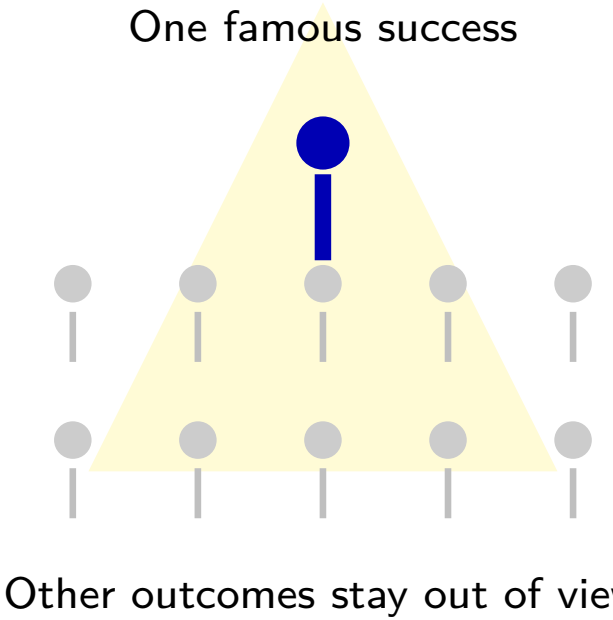
Key point: Confidence is not proof of skill. Expertise can bring both confidence and awareness of limits.

Design implication

Provide practice, clear criteria, and feedback. Test actual task performance instead of relying only on self-rated confidence.

7. Survivorship bias

We can reach misleading conclusions when we only see the successes.



"Successful founders dropped out, so dropping out must help you succeed."

This ignores people who made the same choice and did not succeed.

Ask: Whose experience is missing?

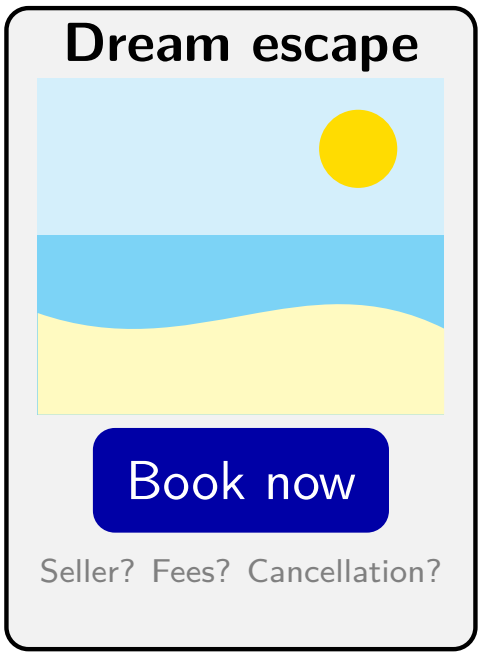
Design implication

Include people who abandoned your app or failed to finish a task. Interviewing only active users misses important problems.

Conceptual illustration, not counts. Success stories alone do not establish cause.

8. Affect heuristic

We use feelings as a shortcut when judging benefits and risks.



You feel excited about a holiday and like the booking app.

You assume the booking is safe before checking the seller or cancellation terms.

Ask: What evidence supports that feeling?

Design implication

Make fees, cancellation terms, and seller information easy to check. Attractive presentation should support informed decisions.

Slovic et al. (2007).

Human decisions are rarely rational



Human decisions are rarely rational



Design Implications

- **Prioritize** your information. People only have around a window of **5-8s**.
- Remind **loose ends** - e.g., return to default mode, disconnect after inactivity, end with action, etc.
- **Convention** and **feedback** helps learning.
- If it is something **new**, make sure learning is fun with **low risk** and **high rewards**.
- Don't assume people **will think or read or learn**. We are irrational.
- Test regularly and quantitatively. Don't only take **average** performance but take note of the **special** cases.

Activities: Investigate one SIS task

Visit `http://sis.ait.ac.th`. Choose one task, such as finding a course or checking a registration deadline.

1. Walk through the task

Note where users might struggle. Use these questions:

- **Perception and memory:** What is hard to notice, read, understand, or remember?
- **Personality and motivation:** Are useful choices available? Are progress and next steps clear?
- **Cognition:** What is hard to learn? Where could users miss information or make a mistaken decision?

2. Choose five improvements

Use **at least three lecture topics** across your five fixes. Choose concepts that explain the problems you find; you do not need to use every topic or add gamification.

Activities: Explain and test your fixes

Submit **five pages, one improvement per page**. State your chosen task at the top of the first page.

1. **Problem:** Annotate a screenshot. Explain what might go wrong during the task.
2. **Concept:** Name a relevant lecture concept and explain how it accounts for the difficulty.
3. **Fix:** Sketch your change and explain why it helps.
4. **Check:** Propose one task and one outcome to compare before and after. No experiment is required.

Example: A deadline is easy to miss

Attention: The deadline blends into unrelated dates. **Fix:** Put it beside the registration action with a clear label. **Check:** Ask users to find the deadline; compare correct answers.

Aim for clear reasoning and useful changes, not the number of theory names.

What's next

In following week, read my slide on **Test** and these complimentary resources:

- Mackenzie, Chapter 6, **Hypothesis Testing**, Human Computer Interaction: An Empirical Research Perspective, 1st ed. (2013)
- Yatani, Advanced Topics in Human-Computer Interaction, `http://yatani.jp/teaching/doku.php?id=2016hci:start`

Please also download **JASP** for our next next week workshop.

Questions